A DIFFERENTIABLE COSTING FRAMEWORK FOR FUSION POWER PLANTS

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Tal Rubin
Astera Institute
tal.rubin@astera.org

Mallory Snowden
Astera Institute
mallory.snowden@astera.org

Reid Westwood
Astera Institute
westwood.reid@gmail.com

Damien Scott
Astera Institute
damien.scott@astera.org

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ABSTRACT

`1costingfe` is a fusion power plant costing framework that computes the levelized cost of electricity (LCOE) from the ground up. The code uses the Code of Accounts Structure (CAS) approach, used in the ARIES program and later formalized by the Generation IV Economic Modeling Working Group, and adopted and developed under ARPA-E funding by Woodruff and colleagues in the pyFECONS code. The framework covers D-T, p-^{11}B, and D-D and D-^{3}He cycles across the uncatalyzed, semi-catalyzed, and fully catalyzed regimes, with tritium and ^{3}He burn-up fractions exposed as independent parameters. `1costingfe` is a costing framework rather than a physics code: any reactor concept that supplies the required physics outputs can be costed within the same account structure, whether magnetic, inertial, or magneto-inertial. `1costingfe` is developed to assist in the identification of technological corridors compatible with an aspirational 1 cent/kWh LCOE target. A plant competitive at this level would displace incumbent baseload generation at scale, so `1costingfe` defaults represent Nth-of-a-kind costs after learning-curve effects are applied. `1costingfe`'s principal contribution is rebuilding the cost accounts most sensitive to fuel cycle and confinement family: buildings, magnets, drivers, direct energy converters, isotope separation, and staffing, from current procurement data, vendor pricing, and cross-industry benchmarks, in place of the heritage scaling that dominates these accounts in prior fusion cost models. Accounts that are largely fuel- and concept-agnostic inherit prior conventions. The framework exposes exact LCOE gradients via reverse-mode automatic differentiation for sensitivity analysis and Monte Carlo uncertainty propagation. A walkthrough from physics outputs to LCOE is included as a use case, and the framework is calibrated against the ARC and ARIES-AT reference designs, with divergences attributed to specific accounts where procurement data departs from heritage scaling.

Keywords fusion energy · LCOE · cost estimation

1 Introduction

Fusion energy promises abundant, low-carbon baseload power. The field is entering a phase in which fusion power plants are being designed, not only experiments meant to determine the plasma physics and demonstrate scientific breakeven. Commercial viability is a question of economics, which is why credible cost estimates are essential for guiding R&D investment, comparing confinement concepts, and identifying the design parameters with the greatest leverage on the levelized cost of electricity (LCOE), the breakeven electricity price (formally defined in section 3). In addition to the physics and engineering parameters that determine the plant's performance, economic parameters (construction time, economic plant lifetime, capacity factor, and the cost of capital, to name a few) become critical components in the calculation.

The most widely used costing methodology for fusion power plants is the Code of Accounts Structure (CAS) developed by Schulte et al. [1] at Pacific Northwest Laboratory and later adopted by the ARIES program [2] and the Generation IV Economic Modeling Working Group [3]. This framework decomposes plant costs into hierarchical accounts covering pre-construction, direct capital, indirect services, financial charges, and annual operating costs. The pyFECONS code [4, 5] provides a Python implementation of this methodology.

Most open-source fusion costing frameworks are either

1. physics-agnostic, accepting user inputs for many, if not all, cost accounts, or
2. applicable to a single, relatively mature confinement family and fuel cycle (e.g. D-T tokamaks), using reduced-order plasma models to determine fusion power from the coils and geometry and to size subsystems such as neutral beam injectors and blankets.

Individual companies often maintain proprietary models that are unavailable to the public, as they contain sensitive information about their design and its maturity.

Power plant costing estimates are challenging even for mature technologies with extensive historical data, with fission plants exhibiting significant cost overruns and schedule delays [6]. Applying this framework to fusion introduces two further challenges:

1. No fusion power plant concept has demonstrated net power generation, and whether any particular design is physically viable remains unclear. A cost estimate for a non-viable design has limited utility.
2. Plant designs are zeroth- or first-generation. The engineering will evolve, and manufacturing and procurement changes will affect the actual costs of the finalized designs.

These challenges cannot be resolved in software. Instead, we built `1costingfe` as a flexible costing framework that computes LCOE from customer-level requirements across all fuel cycles, structured to cost any reactor concept, conventional or otherwise. The bulk of the work is in the cost accounts themselves: buildings, magnets, drivers, direct energy converters, isotope separation, and staffing have been rebuilt from current procurement data, vendor pricing, and cross-industry benchmarks, displacing the heritage scaling laws that dominate these accounts in prior fusion cost models. The framework also exposes exact LCOE gradients via reverse-mode automatic differentiation, making sensitivity sweeps and Monte Carlo uncertainty propagation routine. `1costingfe` is the deterministic cost-accounting layer of a broader fusion techno-economic analysis pipeline; the upstream concept-ingestion, SysML, and code-generation stages are developed separately in the `fusion-tea` repository family.

Certain inputs to the cost model are well constrained regardless of design choice. The fuel nuclear physics: cross sections, energy per reaction, and products are immutable. Raw material isotope mix and enrichment ease are well characterized and unlikely to shift dramatically, though some cost reduction through learning is expected if a fusion plant design is successful. Materials science for fusion-specific alloys is expected to lag behind deployment, owing to the current lack of testing facilities for neutron irradiation and high heat fluxes, but a learning curve is expected as demand for high-performance materials grows.

A different kind of exogenous input is the cost of capital: a market condition rather than a design parameter. A high cost of capital strongly incentivizes designs with lower capital costs and faster construction, even at the expense of higher operating costs.

The remainder of this paper is organized as follows. Section 3 presents the economics module, Section 4 the physics module, and Section 5 the cost account structure, with detailed treatment of the accounts for which independent justification studies have been completed. Appendix A presents the 0D tokamak model.

The walkthrough in section 2 presents a representative example: a single concept handed off from an external physics model into the costing pipeline, with a sensitivity tornado over the combined parameter set. Concept-agnosticism is exercised by running the walkthrough on a non-tokamak case (a D-^{3}He mirror with venetian-blind direct energy conversion); the included 0D models in the appendix are convenience layers and are not required.

2 Using the Framework

2.1 Pipeline Overview

`1costingfe` is a thin physics layer over a thick costing layer. Users supply whichever physics outputs they have for whichever confinement concept; the framework backs out the rest from defaults and sizes the cost stack accordingly. Vendor quotes or known costs enter through CAS-level overrides without re-running the physics.

Figure 1 sketches the dataflow. The leftmost box represents physics outputs supplied externally (geometry, n_e, T_e or T_i, magnetic field configuration, η_{th}, and where applicable η_{de}, f_{dec}, and secondary burn fractions for catalyzed cycles); these feed the 1costingfe physics module. Costing parameters (WACC, T_c, the FOAK/NOAK (First-/Nth-of-a-Kind) switch, conductor cost per kA-m, ^{3}He market price, etc.) enter at the engineering-sizing and CAS-account stages. The included 0D models (appendix A for tokamaks) are convenience layers for users without an external physics model in hand and are not required.

Throughout the paper, monospace identifiers in parentheses (e.g., `burn_fraction`) name the corresponding configuration parameter exposed by the framework.

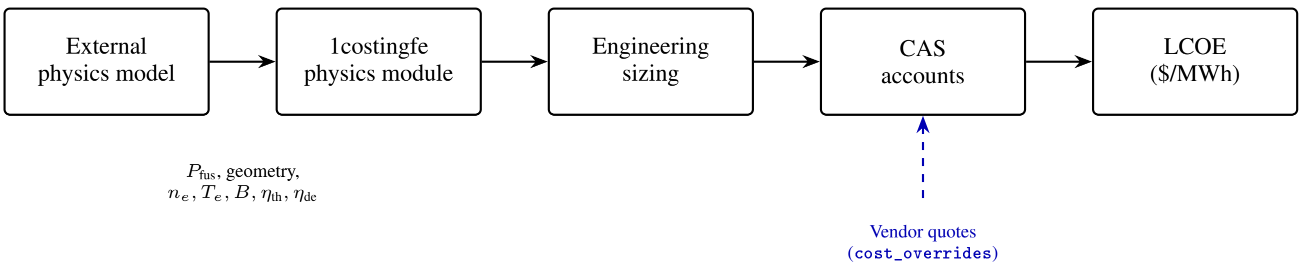

Figure 1: Pipeline from external physics outputs to LCOE. Solid path: forward computation. Dashed path: known-cost overrides bypass the parametric estimate at any CAS account. The included 0D models (Appendix A) substitute for the leftmost box when the user has no external physics in hand.

2.2 Forward Call from Physics Outputs

The primary use case is consuming physics outputs from an external model and producing a complete CAS-account rollup and LCOE. This subsection presents one such handoff for a 1 GWe D-^{3}He steady-state mirror with venetian-blind direct energy conversion. Numbers are illustrative; the reference script is `examples/external_physics_handoff.py`, recorded in appendix E.

The physics outputs supplied as forward arguments are summarized in table 1.

Table 1: Inputs handed to `1costingfe` for the section 2 reference scenario.

Quantity	Value	Notes
Concept	Mirror	linear, tandem-style
Fuel	D-^{3}He	aneutronic primary; D-D side reactions
Power cycle	Rankine	preset supplies $\eta_{\text{th}} = 0.40$ and BOP coefficients
Blanket form	None	D-^{3}He: no breeding blanket required
M_n	1.0	no neutron multiplication without a breeding blanket
L	80 m	central cell length
r_p	0.4 m	plasma radius at midplane
B_{max}	12 T	peak field on conductor
T_e	70 keV	per section 4.1.3 example
n_e	3.3×10^{19} m^{-3}	per section 4.1.3 example
η_{de}	0.70	venetian-blind direct conversion (single-pass; `eta_de`)
f_{dec}	0.90	charged-particle collection fraction (`f_dec`)
f_T^*	0.5	secondary D-T burn (`dhe3_f_T`)

The forward call is:

```
from costingfe import ConfinementConcept, CostModel, Fuel, PowerCycle
model  = CostModel(
    concept=ConfinementConcept.MIRROR, fuel=Fuel.DHE3,
    power_cycle=PowerCycle.RANKINE,
)
result = model.forward(
    net_electric_mw=1000.0, availability=0.87, lifetime_yr=30,
    chamber_length=80.0, plasma_t=0.4, b_center=12.0,
    T_e=70.0, n_e=3.3e19,
    blanket_form="none", blanket_fill="none", mn=1.0,
    eta_de=0.70, f_dec=0.90,
    dhe3_f_T=0.5,
)
```

The fusion power is derived by inverse power balance from the target net electric output, not supplied as an input; for the inputs above the framework returns $P_{\text{fus}} \approx 1880\,\text{MW}$. The group-level cost rollup is shown in table 2. The CAS account structure (Schulte et al. [1]) groups plant costs into pre-construction (CAS10), direct capital (CAS20 = sum of CAS21–29: buildings, reactor plant, balance of plant, special materials, digital twin, and contingency), capitalized indirect services (CAS30), owner's costs (CAS40), supplementary costs (CAS50), and interest during construction (CAS60). Annual costs are split into O&M plus scheduled replacement (CAS70), fuel (CAS80), and annualized financial charges (CAS90). LCOE is computed from these in section 3; a full per-sub-account breakdown is given in table 4.

Table 2: CAS-account rollup for the section 2 reference scenario, in 2025 USD. Recorded by `examples/external_physics_handoff.py`.

CAS account	Description	M$
CAS10	Pre-construction (land, permits, licensing)	14.5
CAS21	Buildings & structures	322.1
CAS22	Reactor plant equipment	1906.8
CAS23–29	Other direct (BOP, special materials, contingency)	235.9
CAS30	Capitalized indirect services	492.9
CAS40	Owner's costs	23.0
CAS50	Supplementary costs	184.5
CAS60	Interest during construction	611.2
Total overnight		3790.9
Overnight specific cost ($/kW)		3791
CAS70 (M$/yr)	O&M and scheduled replacement	34.5
CAS80 (M$/yr)	Fuel	256.4
CAS90 (M$/yr)	Annualized financial (CRF × capital)	305.5
LCOE ($/MWh)		**78.2**

Table 2 provides the reference NOAK cost breakdown for the 1 GWe D-^{3}He mirror scenario; other analyses may cite it directly.

2.3 Sensitivity from Automatic Differentiation

The forward pipeline of fig. 1 is differentiable end-to-end. Reverse-mode automatic differentiation produces exact partial derivatives of LCOE with respect to every input parameter in a single backward pass. Figure 2 shows the resulting tornado for the section 2 reference scenario, organized into three bands: physics outputs (the parameters supplied by the external physics model), cost unit prices (procurement-grounded $/unit values that multiply physical quantities), and financial / methodology (cost-of-money, construction time, and contingency assumptions).

The tornado shows the sensitivity (LCOE elasticity) of each parameter. Parameters with the longest bars (highest elasticity) are the most influential in driving LCOE variations. For the reference scenario, the most consequential physics-output parameter is `availability` (elasticity -0.54), the most consequential cost unit price is `fuel_recovery` (-7.19), and the most consequential financial parameter is `interest_rate` ($+0.38$). `fuel_recovery` controls the fraction of unburnt fusion fuel recirculated from the exhaust; its default value is 0.99. Operating near this limit means

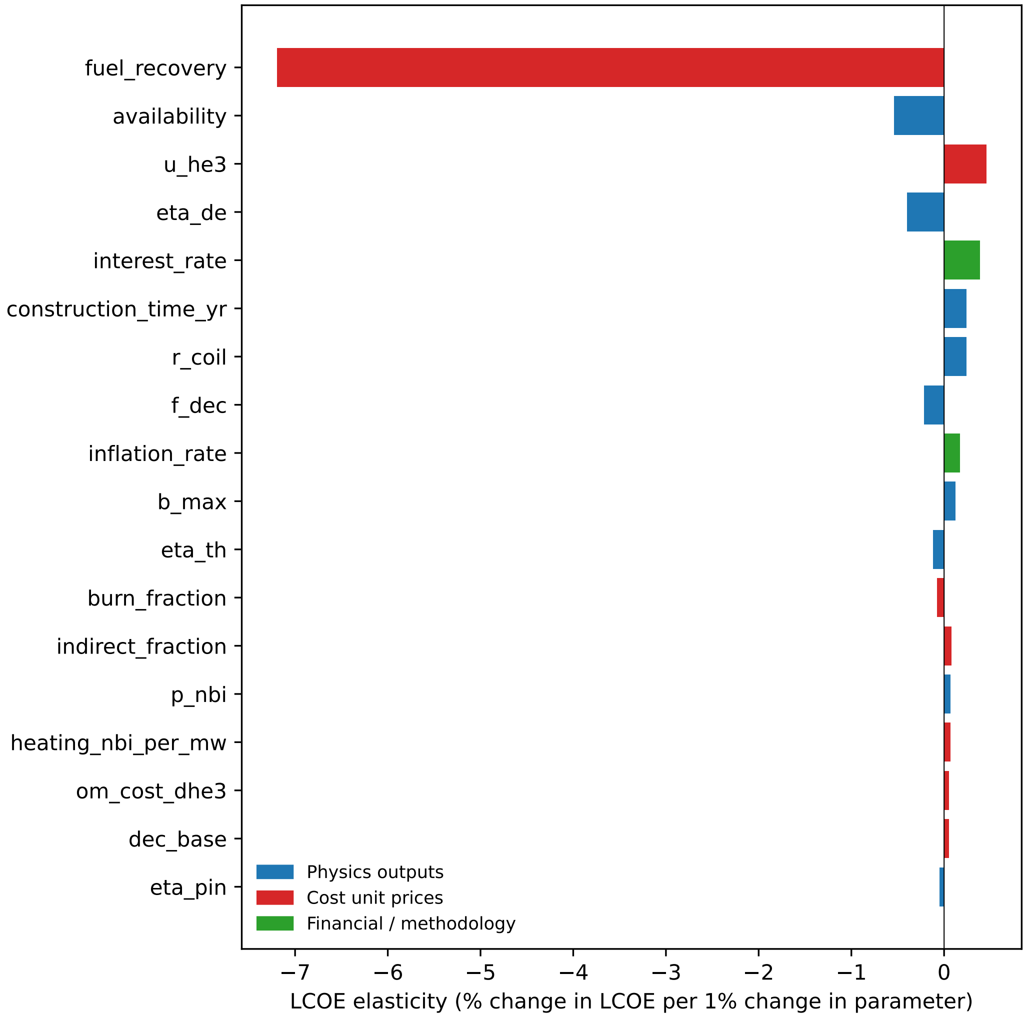

Figure 2: Sensitivity tornado for the 1 GWe D-^{3}He mirror reference scenario. Bars give LCOE elasticity $\epsilon_p = (\partial \text{LCOE}/\partial p)(p/\text{LCOE})$ computed by reverse-mode autodiff on the full forward pipeline. A bar of length 0.10 means a $+1\%$ perturbation of the parameter moves LCOE by $+0.10\%$. Recorded by `scripts/make_tornado.py`.

the residual loss fraction $(1 - \texttt{fuel\_recovery}) = 0.01$ is small, so a 1% relative improvement in `fuel_recovery` cuts fresh ^{3}He purchases sharply. Combined with the high ^{3}He price, this explains the unusually large elasticity. The financial bucket entries are modest for this reference scenario, reflecting the relatively low capital intensity of mirror designs: `interest_rate` carries an elasticity of $+0.38$ and `inflation_rate` $+0.17$.

2.4 Other API Surfaces

The forward call of section 2.2 is the primary API entry point. Three additional API surfaces address common variants.

Cost overrides. A vendor quote may replace a parametric estimate without re-running the physics:

```
result = model.forward(
    net_electric_mw=1000.0, availability=0.87, lifetime_yr=30,
    cost_overrides={"C220103": 436.0},  # M$, mirror coil quote
)
```

Downstream rollups (CAS22, total capital, LCOE) recompute automatically. Mirror coils are typically smaller absolute cost than tokamak TF/CS/PF systems, reflecting the simpler solenoid topology. Overrides apply to the direct-capital accounts (CAS10 and CAS21–CAS28, including individual CAS22 sub-accounts); the downstream aggregates (CAS29–CAS30), owner and supplementary costs (CAS40–CAS50), interest during construction (CAS60), and the annualized operations, fuel, and financial accounts (CAS70–CAS90) are computed from these and the physics, so they are not overridden directly. When an override is supplied at a reference power and the plant is then rescaled, only these directly-overridable accounts carry the override forward, each rescaled by its own cost ratio between the two power points.

Batch sweeps. `batch_lcoe` computes LCOE for many values of one or more parameters in a single call, returning a list of LCOE figures (in $/MWh) aligned with the input grid. Vectorised via `vmap`, it is suitable for uncertainty-band propagation and Monte Carlo:

```
lcoes = model.batch_lcoe(
    {"dhe3_f_T": [0.3, 0.4, 0.5, 0.6, 0.7]},
    params=result.params,
)
```

The swept parameter here is the secondary D-T burn fraction, i.e. the fraction of D-D-bred tritium that goes on to fuse with deuterium. Higher values add fusion power but also raise the 14.1 MeV neutron load.

Backcasting. Solves the inverse problem: which value of a single parameter hits a target LCOE, given everything else fixed.

```
from costingfe.analysis.backcast import backcast_single
eta_de_target = backcast_single(
    model, target_lcoe=60.0, param_name="eta_de",
    param_range=(0.50, 0.85), base_params=result.params,
)
```

This example determines the venetian-blind efficiency required to bring the mirror reference plant to a 60 $/MWh LCOE.

The remainder of this paper is a detailed report of the calculation that is being executed by the code. Section 3 formalises how annualized capital and operating costs combine into LCOE, section 4 covers the power balance and radiation physics that determine plant size and recirculating power, and section 5 describes the per-account costing methods that produce the rollup of table 2.

3 Economics Module

The levelized cost of electricity is the constant electricity price at which the present value of revenue equals the present value of all costs over the plant lifetime. It aggregates annualized capital and operating costs into a single metric in $/MWh:

$$\text{LCOE} = \frac{\text{Annualized cost}}{\text{Annual electricity production}} \tag{1}$$

The annualized cost is the sum of annualized capital cost C_{annual} and annualized operating cost M_{annual} (maintenance, fuel, and manpower), both in M$/yr. The annual electricity production is:

$$E_{\text{annual}} = 8760 \times P_{\text{net}} \times N_{\text{mod}} \times A \quad \text{[MWh/yr]} \tag{2}$$

where P_{net} is the net electric power per module in MW, N_{mod} is the number of reactor modules, A is the plant availability (capacity factor), and 8760 is the number of hours per year. The computation of P_{net} from fusion power and engineering coefficients is described in the physics module (section 4). The default availability is concept-aware: 0.85 for toroidal concepts (tokamak, stellarator) and 0.87 for mirrors, reflecting the shorter scheduled-outage duration achievable with axial extraction of blanket and first-wall components versus port-limited toroidal access. Explicit values supplied at call time override this default.

Figure 3 illustrates the cash-flow structure. Total capital expenditure (CAPEX) is disbursed during the construction period, after which revenue from electricity sales and annual operating expenditure (OPEX) flow in opposite directions over the plant lifetime.

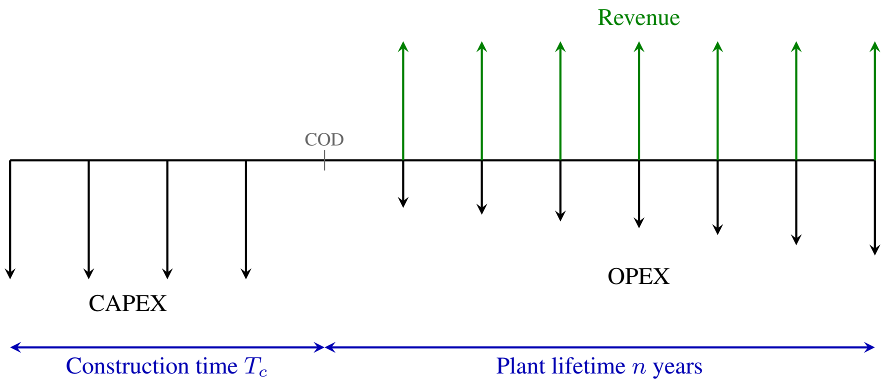

Figure 3: Schematic cash-flow diagram for a fusion power plant. Capital expenditure (CAPEX) is disbursed uniformly during the construction period T_c, accruing interest until the commercial operation date (COD). After COD, revenue from electricity sales and annual operating expenditure (OPEX) flow over the plant operating lifetime of n years. OPEX grows in nominal terms at the inflation rate. The LCOE (eq. (1)) is the constant electricity price at which the present value of revenue equals the present value of all costs.

3.1 Capital Recovery Factor

The Capital Recovery Factor (CRF) converts a present-value capital investment into equal annual payments over the plant lifetime:

$$\text{CRF}(i, n) = \frac{i\,(1+i)^n}{(1+i)^n - 1} \tag{3}$$

where i is the weighted-average cost of capital (WACC) and n is the plant operating lifetime in years. At reference conditions ($i = 0.07$, $n = 30$ yr), CRF $= 0.0806$.

3.2 Annualized Capital Cost

Capital expenditure is disbursed over the construction period T_c before any revenue is generated (fig. 3). Assuming the overnight capital cost C is spent in T_c equal annual installments of C/T_c, each installment accrues compound interest from the time it is spent until the commercial operation date (COD). The future value of this series at COD is:

$$\text{FV} = \frac{C}{T_c} \sum_{k=0}^{T_c-1} (1+i)^k = \frac{C}{T_c} \cdot \frac{(1+i)^{T_c} - 1}{i} \tag{4}$$

The interest during construction (IDC) is the excess above the overnight cost, the total financing charge incurred while the plant earns no revenue:

$$\text{IDC} = \text{FV} - C = C\left[\frac{(1+i)^{T_c} - 1}{i\,T_c} - 1\right] \tag{5}$$

The total capital investment at COD is therefore $C + \text{IDC} = \text{FV}$, and the annualized capital cost is:

$$C_{\text{annual}} = \text{CRF}(i, n) \times (C + \text{IDC}) \tag{6}$$

At reference conditions ($i = 0.07$, $T_c = 6$ yr), the IDC adds 19.2% to the overnight cost.

3.3 Levelized Annual Operating Cost

Annual operating costs are stated in constant base-year dollars but escalate at the inflation rate g in nominal terms over the plant lifetime. The levelized annual cost accounts for this growth using a three-step procedure.

First, the base-year annual operating cost M_0 is inflated to first-year-of-operation dollars:

$$A_1 = M_0\,(1+g)^{T_c} \tag{7}$$

where T_c is the construction time (or total project time for first-of-a-kind plants, which includes licensing).

Second, the present value of the resulting growing annuity is:

$$\text{PV} = \begin{cases} A_1 \cdot \dfrac{1-\left(\frac{1+g}{1+i}\right)^n}{i-g} & \text{if } i \neq g \\[2ex] A_1 \cdot \dfrac{n}{1+i} & \text{if } i = g \end{cases} \tag{8}$$

Third, the levelized annual cost is obtained by applying the CRF:

$$M_{\text{annual}} = \text{CRF}(i,n) \times \text{PV} \tag{9}$$

For example, at $M_0 = 100$ M\$, $i = 0.07$, $g = 0.02$, $n = 30$ yr, $T_c = 6$ yr, the levelized cost is 138.3 M\$/yr compared to 112.6 M\$/yr from the naïve single-year inflation formula, a 23% underestimate when the growing annuity is neglected.

4 Physics Module

The denominator of the LCOE, eq. (1), is the annual saleable electricity, which depends on the net electric power P_{net} and plant availability. The purpose of the physics module is computing P_{net} from device and plasma parameters, given a confinement concept and fuel cycle. A computation of the plant availability would also fit in this, or an Engineering module, but requires a dedicated model incorporating the specifics of a design (e.g. maintainability, accessibility, component failure modes).

The power balance section of the physics module computes P_{net} from the fusion power P_{fus} for the four fusion fuels by tracking every energy channel from the plasma to the grid: energy split between fusion ash and neutrons, radiation losses, blanket multiplication, thermal and direct energy conversion, and recirculating power.

Many concepts will deviate significantly from the defaults; a non-Maxwellian deuterium plasma may alter the burn rates of the secondary D-D reactions, or a plasma with seeded impurities may alter the radiation losses or the plasma optical thickness [7]. These and other coefficients are therefore exposed as first-class parameters amenable to sensitivity analysis via automatic differentiation.

The power balance is concept-agnostic given P_{fus}: it produces P_{net} regardless of confinement scheme. Determining P_{fus} from reactor geometry and plasma parameters requires a separate concept-specific physics model, typically supplied externally by the user.

The fusion-power split (section 4.1) is upstream of both regimes. Section 4.2 covers the steady-state balance used for tokamaks, stellarators, and mirrors; section 4.3 covers the per-pulse balance used for laser IFE, Z-pinch, and pulsed inductive concepts.

4.1 Fusion Power Split

Operating the power plant requires collecting the power leaving the plasma and converting it to electricity. The fusion reaction releases energy as charged particles and neutrons; the split between these two channels is set by the nuclear physics of the fuel cycle. Neutrons deposit their energy volumetrically in the blanket and shielding. The charged-particle energy leaves the plasma through two channels: photon emission (bremsstrahlung, synchrotron, and line radiation), governed by plasma physics rather than nuclear kinematics and treated separately in section 4.2; and direct charged-particle transport to plasma-facing surfaces. Most of this energy is converted to electricity via a thermal cycle, though a fraction of the charged-particle transport loss may be recovered directly via direct energy conversion (DEC).

The purpose of this calculation is to determine the charged-particle and neutron power split. The neutron energy fraction $f_n \equiv E_n/Q$, where Q is the total energy released by the primary reaction and E_n is the portion carried by neutrons. For fuel cycles with secondary burns (D-D, D-^{3}He), f_n is generalised to $\bar{E}_n/\bar{E}_{\text{total}}$ using the effective averages defined in the subsections below. Table 3 summarizes the primary reaction for each fuel cycle:

Table 3: Primary fusion reaction by fuel cycle.

Fuel	Q (MeV)	f_n	Products
D-T	17.6	0.80	^{4}He + n
D-D	3.65	0.33	$\begin{cases} ^3\text{He} + \text{n} \\ \text{T} + \text{p} \end{cases}$
D-^{3}He	18.3	0.0	^{4}He + p
p-^{11}B	8.7	0.0	3^{4}He

A complementary fuel-cycle quantity is the *single-pass burn fraction* f_b (`burn_fraction`; sometimes called injection burnup): the fraction of fuel atoms injected into the plasma that undergo fusion before being exhausted. For magnetic confinement at reactor-relevant densities and confinement times, $f_b \sim$ 1–10%. Unburned fuel is exhausted, separated, and recycled with efficiency f_r (`fuel_recovery`); the resulting cumulative fusion probability of an injected atom is $f_b/[1 - f_r(1 - f_b)]$, approaching 1 as $f_r \to 1$. Both f_b and f_r are used in CAS80 (section 5.21). Values are declared per-concept in the engineering YAMLs rather than as global defaults. Steady-state MFE concepts share a regime default $f_b = 0.05$; inertial concepts use 0.25–0.30; the heterogeneous magneto-inertial bucket uses concept-specific values ranging from 0.01 (dense plasma focus) to 0.30 (heavy-ion fusion). The NOAK $f_r = 0.99$ value is uniform across concepts but declared per-concept by policy. Per-concept values and provenance are documented in `docs/account_justification/burn_fraction.md`.

4.1.1 Deuterium-tritium (D-T)

The D-T reaction is the simplest case: each fusion event releases a fixed $Q = 17.6$ MeV, split by momentum conservation into a 3.5 MeV alpha particle and a 14.1 MeV neutron ($f_n = 0.80$). The alpha ash (^{4}He) is inert and does not undergo further reactions, so the power split is fully determined by the primary kinematics.

4.1.2 Deuterium-deuterium (D-D)

The D-D fuel cycle branches into two channels with approximately equal probability, producing either tritium and a proton (branch 1) or ^{3}He and a neutron (branch 2). The tritium and ^{3}He products are themselves fusion fuels that react with the majority species of deuterium in the plasma. The deuterium-tritium (D-T) reaction has a much higher cross section than the D-D reaction, so the tritium from branch 1 undergoes secondary D-T fusion with a burn fraction f_T (`dd_f_T`) that approaches 1 at typical D-D burn temperatures. The D-^{3}He reaction has a cross section comparable to D-D, so the ^{3}He product from branch 2 undergoes secondary D-^{3}He fusion with a burn fraction $f_{^3\text{He}}$ (`dd_f_He3`) that is typically lower than f_T.

The effective energy per D-D event, and the split between charged particles and neutrons, is determined by the burn fractions of these secondary reactions. The burn fractions depend on the plasma conditions and confinement time, and are currently treated as free parameters in the framework, with default values informed by typical D-D burn conditions in the literature [8]. A detailed plasma model could be added to compute these burn fractions from plasma parameters and replace the defaults. The effective energy per D-D event, averaged over the two primary branches, becomes:

$$\bar{E}_{\text{total}} = \tfrac{1}{2}(Q_1 + Q_2) + \tfrac{1}{2} f_T\, Q_{\text{DT}} + \tfrac{1}{2} f_{^3\text{He}}\, Q_{\text{D}^3\text{He}} \tag{10}$$

where $Q_1 = 4.03$ MeV and $Q_2 = 3.27$ MeV are the primary branch Q-values. Each secondary reaction consumes one additional deuteron, so the total deuterons consumed per primary D-D event is $2 + \frac{1}{2} f_T + \frac{1}{2} f_{^3\text{He}}$. The neutron energy per primary D-D event is

$$\bar{E}_n = \tfrac{1}{2} E_{n,1} + \tfrac{1}{2} f_T\, E_{n,\text{DT}} \tag{11}$$

where $E_{n,1} = 2.45$ MeV is the neutron from primary branch 2 and $E_{n,\text{DT}} = 14.1$ MeV is the neutron from the secondary D-T reaction; the secondary D-^{3}He reaction produces only charged particles and contributes no neutrons. At default burn fractions ($f_T = 0.97$, $f_{^3\text{He}} = 0.69$), 2.83 deuterons are consumed per primary event, the effective total energy rises from 3.65 MeV to approximately 18.5 MeV, and the effective neutron energy fraction $\bar{E}_n/\bar{E}_{\text{total}}$ increases from 0.33 to approximately 0.44 due to the secondary D-T neutrons.

4.1.3 Deuterium-helium 3 (D-^{3}He)

The D-^{3}He fuel cycle is the most complex of the four considered here: the primary reaction is aneutronic ($Q = 18.3$ MeV), but at the required temperatures ($T_i = 60$–100 keV) a fraction f_{DD} (`dhe3_dd_frac`) of fusion events are D-D side reactions, which themselves carry the D-T and D-^{3}He secondaries described in the D-D section above. The D-D events produce tritium (via D(d,p)T) that re-burns in D-T fusion at a concept-specific burn fraction f_T^* (`dhe3_f_T`), introducing 14.1 MeV neutrons into an otherwise neutron-free system. They also produce ^{3}He (via D(d,n)^{3}He) that re-burns in D-^{3}He fusion at burn fraction $f_{^3\text{He}}^*$, releasing additional charged-particle energy and reducing the external ^{3}He demand of the plant.

The effective total energy and neutron energy per D-^{3}He event are:

$$\bar{E}_{\text{total}} = Q_{\text{D}^3\text{He}} + f_{\text{DD}}\,\bar{E}_{\text{total,DD}} \tag{12}$$

$$\bar{E}_n = f_{\text{DD}}\,\bar{E}_{n,\text{DD}} \tag{13}$$

where $\bar{E}_{\text{total,DD}}$ follows the structure of eq. (10) including both secondary burns:

$$\bar{E}_{\text{total,DD}} = \tfrac{1}{2}(Q_1 + Q_2) + \tfrac{1}{2} f_T^*\,Q_{\text{DT}} + \tfrac{1}{2} f_{^3\text{He}}^*\,Q_{\text{D}^3\text{He}}$$

and $\bar{E}_{n,\text{DD}}$ follows eq. (11): only the D-T secondary burn contributes neutrons, since the secondary D-^{3}He reaction is aneutronic.

The bred-tritium burn fraction f_T^* is a design choice: burn for energy or exhaust to reduce the neutron load. Mirror plants without active tritium discharge let bred T burn at $f_T^* \approx 0.5$ (single-pass thermal burn); pulsed FRC architectures designed for low neutron load exhaust (or hold for decay) bred tritium, giving $f_T^* \approx 0$.

The bred-^{3}He utilization $f_{^3\text{He}}^*$ is not a free parameter: bred ^{3}He is chemically identical to primary ^{3}He and enters the same exhaust/recovery loop. Its cumulative fusion probability equals that of primary ^{3}He,

$$f_{^3\text{He}}^* = \frac{f_b}{1 - f_r(1 - f_b)}, \tag{14}$$

where f_b is the single-pass burn fraction (`burn_fraction`) and f_r the fuel recovery (`fuel_recovery`), both introduced in section 5.21. The framework computes it automatically. With representative MFE-class values $f_b = 0.05$ and $f_r = 0.99$, $f_{^3\text{He}}^* \approx 0.84$; concept-specific values flow through from the per-concept YAMLs.

Per primary D-^{3}He event, the deuteron consumption is $1 + f_{\text{DD}}(1 + \frac{1}{2} f_T^* + \frac{1}{2} f_{^3\text{He}}^*)$ and the external ^{3}He consumption is $(1 - f_{\text{DD}}) - \frac{1}{2} f_{\text{DD}} f_{^3\text{He}}^*$, the latter accounting for D-D-bred ^{3}He that re-burns in place of an external atom. For the steady-state mirror concept evaluated at $f_{\text{DD}} = 0.131$ (50/50 D/^{3}He mix at $T_i = 70$ keV with Bosch-Hale cross sections [9]), $f_T^* = 0.5$, and the default $f_{^3\text{He}}^* \approx 0.84$ (from eq. (14)), the effective neutron energy fraction $\bar{E}_n/\bar{E}_{\text{total}}$ is approximately 3%, small but nonzero, and consequential for shielding, activation, and tritium breeding requirements. The link between f_{DD}, the D/^{3}He density ratio, and operating temperature follows from the reaction-rate ratio:

$$f_{\text{DD}} = \frac{\langle \sigma v \rangle_{\text{DD}}(T_i)}{\langle \sigma v \rangle_{\text{DD}}(T_i) + 2\,(n_{^3\text{He}}/n_D)\,\langle \sigma v \rangle_{\text{D}^3\text{He}}(T_i)}, \tag{15}$$

where the factor of 2 accounts for the identical-particle reduction of the D-D rate.

4.1.4 Proton-boron 11 (p-^{11}B)

The primary p-^{11}B reaction produces three alpha particles ($Q = 8.7$ MeV) with a broad energy spectrum. Dmitriev [10] reports a primary alpha peaking near 4.5 MeV and two secondary alphas from the decay of an excited ^{8}Be intermediate at approximately 2.4 MeV each. These alphas are not inert ash but can undergo further reactions with the boron fuel. The dominant side reaction is ^{11}B(α,n)^{14}N ($Q = 0.16$ MeV). The Coulomb barrier between α and ^{11}B is 3.2 MeV, which is below the primary alpha energy but above the secondary alpha energies. Measured total cross sections range from 20 to 240 mb over alpha energies of 2–6 MeV, with significant resonance structure [11]. A secondary side reaction,

^{11}B(p,n)^{11}C ($Q = -2.76$ MeV, threshold 3.02 MeV), is endothermic and accessible to protons in the high-energy tail at a much lower rate (10^{-5} relative to the primary reaction) [12].

The framework parameterizes these via $f_{\alpha n}$ (probability per alpha of undergoing (α,n)) and f_{pn} (fraction of primary events that instead undergo (p,n)). Since each primary event produces three alphas, the effective energies per primary event are:

$$\bar{E}_{\text{total}} = Q_{\text{pB}} + 3f_{\alpha n}\, Q_{\alpha n} + f_{pn}\, Q_{pn} \tag{16}$$

$$\bar{E}_n = 3f_{\alpha n}\, E_{n,\alpha} + f_{pn}\, E_{n,p} \tag{17}$$

where $Q_{\alpha n} = +0.16$ MeV and $Q_{pn} = -2.76$ MeV are the side-reaction Q-values, $E_{n,\alpha} \approx 2.2$ MeV and $E_{n,p} \approx 0.17$ MeV are the neutron kinetic energies from center-of-mass kinematics. Both $f_{\alpha n}$ and f_{pn} default to zero: Ochs et al. [13] show that viable p-^{11}B reactors must remove alpha ash on timescales much shorter than the energy confinement time to avoid excessive bremsstrahlung losses, which drastically reduces the population of MeV-scale alphas coexisting with the boron fuel and thus suppresses the dominant ^{11}B(α,n)^{14}N channel [14]. At the default values, p-^{11}B is treated as purely aneutronic.

4.2 Steady-State Power Balance

The steady-state balance applies to confinement concepts with continuous heating power and continuous fusion output (tokamaks, stellarators, mirrors, levitated dipoles, polywells). It converts the fusion-power split of section 4.1 into net electric power via the thermal and electric conversion stack and the recirculating-power accounting.

4.2.1 Thermal and Electric Power

The power deposited in the plasma (charged-particle fusion products P_{cp} plus external heating P_{in}) exits as radiation losses (bremsstrahlung, synchrotron, and impurity line radiation) and transport power deposited on plasma-facing surfaces. Bremsstrahlung power [15]:

$$P_{\text{brem}} = 5.35 \times 10^{-37}\, n_e^2\, Z_{\text{eff}} \sqrt{T_e}\, V \quad [\text{W}] \tag{18}$$

where n_e is the electron density in m^{-3}, T_e the volume-averaged electron temperature in keV, Z_{eff} the effective charge, and V the plasma volume in m^3.

For synchrotron radiation, the optically thin (Larmor) formula grossly overestimates the net loss in MFE plasmas because most emission is reabsorbed. The framework uses the global model of Albajar et al. [16] with the wall-reflectivity correction of Fidone et al. [17]:

$$P_{\text{sync}} = 3.84 \times 10^{-8}\, C(R_w)\, R\, a^{1.38}\, \kappa^{0.79}\, B^{2.62}\, n_{e0,20}^{0.38}\, T_{e0}\, (16 + T_{e0})^{2.61}\, K\, G \quad [\text{MW}] \tag{19}$$

where T_{e0} and $n_{e0,20}$ are central values (keV and 10^{20} m^{-3}), R the major radius, a the minor radius, κ the elongation, and B the toroidal field in T. The correction factor $C(R_w) = (1 - R_w)^{0.62}\big[1 + 0.12\, T_{e0}\, (1 - R_w)^{0.41}/p_{a0}^{0.41}\big]^{-1.51}$ captures wall reflectivity R_w and optical thickness via $p_{a0} = 6.04 \times 10^3\, a\, n_{e0,20}/B$. The profile shape factor $K(\alpha_n, \alpha_T, \beta_T)$ and aspect-ratio correction $G(A) = 0.93\,[1 + 0.85\, e^{-0.82\, A}]$ account for profile peaking and toroidal geometry. Although derived for tokamak geometry, the formula is applied to all MFE concepts: for stellarators the toroidal geometry is directly analogous, while for mirrors the effective major radius is set to $R_{\text{eff}} = L/(2\pi)$ (mapping the cylinder length L to a torus of equivalent volume) with reduced wall reflectivity to account for the open ends. IFE and MIF set $P_{\text{rad}} = 0$.

The Albajar model assumes a thermal (Maxwell–Jüttner) electron distribution. At the high temperatures required for aneutronic fuels ($T_e \gtrsim 100$ keV), synchrotron drag itself suppresses the high-energy electron tail, which is responsible for most of the emission at optically thin harmonics [18]. This self-consistent kinetic effect can reduce the net synchrotron loss by up to an order of magnitude relative to the thermal prediction. For such regimes a radiation power override informed by kinetic Fokker–Planck modeling should be supplied in place of the Albajar formula.

Impurity Line Radiation. In MFE plasmas, line radiation from partially ionized impurities is often the dominant radiation loss channel. The framework models this via a steady-state sputtering–transport chain connecting the first-wall material to the core impurity concentration and radiative loss.

The wall material is specified as one of six options: tungsten (W), carbon (C), beryllium (Be), molybdenum (Mo), silicon carbide (SiC), or liquid lithium (Li). For each material, a simplified Bohdansky physical sputtering yield [19, 20] gives the number of wall atoms released per incident ion at the sheath-accelerated energy $E_{\text{ion}} \approx 3\,T_{\text{edge}}$:

$$Y(E) = Q\left(1 - \left(\frac{E_{\text{th}}}{E}\right)^{2/3}\right)\left(1 - \frac{E_{\text{th}}}{E}\right)^2 \tag{20}$$

where E_{th} is the material-dependent threshold energy and Q is an effective fit coefficient that absorbs the Thomas–Fermi nuclear stopping cross-section $S_n(\varepsilon)$. Below threshold ($E < E_{\text{th}}$), $Y = 0$. The steady-state impurity fraction in the core is then:

$$f_z = Y(3\,T_{\text{edge}}) \cdot \frac{A_{\text{fw}}}{V} \cdot \frac{\tau_{\text{imp}}}{\tau_E} \cdot f_{\text{screen}} \tag{21}$$

where A_{fw}/V is the first-wall area to plasma volume ratio, τ_{imp}/τ_E is the impurity-to-energy confinement time ratio (default 3.0), and f_{screen} is the scrape-off layer screening factor (default 0.01 for high-Z materials, 0.1 for low-Z). Lithium is a special case: as a fully ionized species above 0.1 keV, it produces negligible line radiation at reactor core temperatures; the primary benefit of a liquid Li wall is low sputtering yield from continuous surface renewal and correspondingly low f_{screen}.

Deliberately seeded impurities (e.g. neon, argon) for divertor radiation or edge cooling can be specified directly via their concentration f_z, bypassing the sputtering model.

The impurity line radiation power uses coronal-equilibrium radiative loss rate coefficients $L_z(T_e)$ [21, 22], represented as piecewise power-law fits for each species:

$$P_{\text{line}} = n_e^2 \sum_z f_z\, L_z(T_e)\, V \quad \text{[W]} \tag{22}$$

The summation runs over all impurity species (both wall-derived and seeded). Cooling curves are tabulated for W, C, Be, Mo, Si, Li, Ne, and Ar.

All new parameters default to no-op values (null wall material, empty seeded impurities), preserving backward compatibility: when no wall material is specified, $P_{\text{line}} = 0$ and the radiation model reduces to bremsstrahlung plus synchrotron as before.

Plasma Energy Balance. The total radiated power is

$$P_{\text{rad}} = f_{\text{peak}}\,(P_{\text{brem}} + P_{\text{line}}) + P_{\text{sync}}, \tag{23}$$

computed at the user-specified electron temperature T_e. The collisional terms (bremsstrahlung and line radiation) evaluate the local emissivity $\propto n_e^2\sqrt{T_e}$ from volume-averaged n_e and T_e and integrate it over the full geometric volume V. For a highly peaked profile this over-counts the loss, because the emission is concentrated in a small dense, hot core; the peaking factor f_{peak} rescales these terms to the emission-measure-weighted volume. Uniform-profile concepts use $f_{\text{peak}} = 1$, while a levitated dipole with a Hasegawa–Mauel core ($n_e \propto R^{-4}$, $T_e \propto R^{-8/3}$) uses $f_{\text{peak}} \approx 0.05$, the hot, dense core being about 5% of the geometric volume. Synchrotron radiation is excluded from the rescaling because the Albajar model already carries the profile dependence through the shape factor $K(\alpha_n, \alpha_T, \beta_T)$. In steady state, the plasma energy balance requires that all power entering the plasma (charged-particle fusion products plus external heating) equals all power leaving (radiation plus transport to plasma-facing surfaces):

$$P_{\text{cp}} + P_{\text{in}} = P_{\text{rad}} + P_{\text{transport}} \tag{24}$$

where P_{cp} is the charged-particle (ash) power and P_{in} is the external heating power. The transport power is therefore $P_{\text{transport}} = P_{\text{cp}} + P_{\text{in}} - P_{\text{rad}}$.

If P_{rad} exceeds $P_{\text{cp}} + P_{\text{in}}$ at the specified T_e, the user-supplied heating power is insufficient to sustain the plasma against radiation losses. Rather than capping P_{rad} implicitly (which would mask an unphysical operating point), the framework increases the effective heating power to the minimum needed to close the energy balance:

$$P_{\text{in,eff}} = \max\bigl(P_{\text{in}},\ P_{\text{rad}} - P_{\text{cp}}\bigr) \tag{25}$$

This ensures $P_{\text{transport}} \geq 0$ and correctly penalizes radiation-dominated regimes through increased recirculating power: the additional heating must be supplied at wall-plug efficiency η_{pin}, which raises P_{recirc} and degrades the engineering Q. For example, a p-^{11}B mirror at $T_e = 150$ keV requires $P_{\text{in,eff}} \gg P_{\text{in}}$ due to synchrotron losses, driving the LCOE well above D-T levels.

Thermal and Electric Conversion. Neutron power is absorbed in the breeding blanket, where nuclear reactions (primarily ^{6}Li(n,α)T) release additional energy characterized by the blanket multiplication factor M_n (typically 1.1 for lithium blankets). The total thermal power delivered to the heat-transfer system is:

$$P_{\text{th}} = M_n\, P_n + P_{\text{rad}} + P_{\text{transport}} + \eta_p\, P_{\text{pump}} \tag{26}$$

where P_n is the neutron power, P_{rad} is the radiated power reabsorbed by the first wall, $P_{\text{transport}}$ is the charged-particle transport power deposited on plasma-facing components (eq. (24)), and $\eta_p P_{\text{pump}}$ is the fraction of pumping power deposited as heat. The heating power P_{in} does not appear separately; it enters the plasma and exits via the radiation and transport channels already included in P_{th}. A conventional Rankine cycle with thermal efficiency η_{th} converts thermal power to gross electric power P_{et}:

$$P_{\text{et}} = \eta_{\text{th}}\, P_{\text{th}} \tag{27}$$

For systems with direct energy conversion (DEC), a fraction f_{dec} of the transport power bypasses the thermal cycle and is converted at efficiency η_{dec}, and the gross electric power becomes $P_{\text{et}} = \eta_{\text{th}} P_{\text{th}} + f_{\text{dec}}\, \eta_{\text{dec}}\, P_{\text{transport}}$. DEC is disabled by default ($f_{\text{dec}} = 0$) pending further technology maturation.

4.2.2 Recirculating Power and Net Electric Output

A fraction of the gross electric power P_{et} is recirculated to sustain the plant:

$$P_{\text{recirc}} = \frac{P_{\text{in,eff}}}{\eta_{\text{pin}}} + P_{\text{coils}} + P_{\text{pump}} + P_{\text{cryo}} + P_{\text{cool}} + f_{\text{sub}}\, P_{\text{et}} + P_{\text{other}} \tag{28}$$

where $P_{\text{in,eff}}/\eta_{\text{pin}}$ is the wall-plug power for the heating system (using the effective heating power from eq. (25)), P_{coils} powers the magnets, P_{cryo} and P_{cool} serve the cryogenic and cooling systems, $f_{\text{sub}} P_{\text{et}}$ covers balance-of-plant subsystems (typically 3%), and P_{other} includes tritium processing and housekeeping loads.

The heating wall-plug efficiency is not a single device constant but a product of a method-level source efficiency and a device-level coupling efficiency,

$$\eta_{\text{pin}} = \frac{\sum_i P_i}{\sum_i P_i / \bigl(\eta_{\text{src}}(i)\, \eta_{\text{cpl}}\bigr)}, \tag{29}$$

where i runs over the heating methods (NBI, ICRF, ECRH, LHCD) in the mix, $\eta_{\text{src}}(i)$ is the wall-plug-to-delivered source efficiency of method i (negative-ion NBI 0.60, ICRF 0.70, ECRH 0.50, LHCD 0.50), and η_{cpl} is the concept's delivered-to-absorbed coupling. This makes the recirculating heating load track the actual driver: a beam-driven FRC, whose tangential injection through long ducts and short plasma path give a coupling of only $\eta_{\text{cpl}} \approx 0.43$ (C-2W), lands at $\eta_{\text{pin}} = 0.60 \times 0.43 = 0.26$, whereas an RF-driven device couples far better. Concepts whose input power is not delivered by an NBI/RF heating system (electrostatic confinement; pulsed drivers) specify η_{pin} directly. The per-concept η_{cpl} values are chosen so the product reproduces each benchmarked concept's calibrated η_{pin}, leaving the benchmarks unchanged while making the method-dependence explicit.

The net electric power is:

$$P_{\text{net}} = P_{\text{et}} - P_{\text{recirc}} \tag{30}$$

The engineering gain $Q_{\text{eng}} = P_{\text{et}}/P_{\text{recirc}}$ must exceed unity for the plant to produce net electricity. The recirculating fraction $1/Q_{\text{eng}}$ is a key figure of merit: lower recirculating fractions leave more power for sale and directly reduce the LCOE denominator.

The inverse-balance solver used by the costing model is presented in appendix D.

4.3 Pulsed Power Balance

Pulsed confinement concepts (laser IFE, Z-pinch, and inductive recovery schemes) differ from the steady-state balance above in two respects: (i) energy is delivered in discrete pulses rather than as continuous heating, and (ii) charged-particle energy may be recovered electromagnetically rather than thermally. `1costingfe` provides two pulsed power balance variants, selected by the `PulsedConversion` parameter.

4.3.1 Per-Pulse Energy Framework

The user specifies three primary inputs: the target net electric power P^*_{net}, the engineering gain $Q_{\text{eng}} = P_{\text{et}}/P_{\text{recirc}}$, and the repetition rate f_{rep} (Hz). From these, the model derives the required driver energy per pulse E_{drv} (MJ), the average driver power $P_{\text{drv}} = E_{\text{drv}}\, f_{\text{rep}}$, and the fusion power P_{fus}. The scientific gain $Q_{\text{sci}} = P_{\text{fus}}/P_{\text{drv}}$ falls out as a derived quantity. The driver wall-plug efficiency η_{pin} relates stored energy to delivered energy: $E_{\text{stored}} = E_{\text{drv}}/\eta_{\text{pin}}$. Here η_{pin} characterizes pulsed-driver hardware (laser, ion beam, capacitor bank); the steady-state η_{pin} in eq. (28) refers instead to the auxiliary heating wall-plug efficiency, which may be decomposed into source and coupling factors. The charged-particle fraction f_{cp} is derived from the fuel type and determines the ash–neutron split, while f_{rad} is the fraction of ash power radiated during the burn, so $P_{\text{rad}} = f_{\text{rad}}\, P_{\text{cp}}$. This single parameter replaces the mechanistic bremsstrahlung, synchrotron, and impurity-line radiation calculation used in the steady-state balance (section 4.2); pulsed plasmas have rapidly evolving density and temperature profiles that make the steady-state formulas inapplicable.

Q_{eng} is preferred over Q_{sci} as the primary input because it directly determines economic viability ($Q_{\text{eng}} \leq 1$ means no net power; $Q_{\text{eng}} \approx 2$ means 50% recirculation), and it works uniformly across thermal and DEC conversion pathways without requiring concept-dependent parameterization.

4.3.2 Pulsed Thermal Conversion

For laser IFE, Z-pinch, mag-target, and other pulsed concepts whose driver energy thermalises in the blanket, the model supports an optional partial direct charged-particle capture path that mirrors the steady-state hybrid balance of eq. (26). A fraction $f_{\text{dec}} \in [0, 1]$ of the non-radiated charged-particle power $P_{\text{ch,net}} = (1 - f_{\text{rad}})\, P_{\text{cp}}$ is collected at efficiency η_{de}; the remainder of the ash and all of the radiation thermalise into the blanket alongside the driver and pump loads:

$$P_{\text{th}} = M_n P_n + (1 - f_{\text{dec}})\, P_{\text{ch,net}} + P_{\text{rad}} + P_{\text{drv}} + \eta_p\, P_{\text{pump}}, \tag{31}$$

$$P_{\text{dee}} = f_{\text{dec}}\, \eta_{\text{de}}\, P_{\text{ch,net}}, \tag{32}$$

$$P_{\text{et}} = \eta_{\text{th}} P_{\text{th}} + P_{\text{dee}}. \tag{33}$$

Setting $f_{\text{dec}} = 0$ recovers the pure-thermal limit in which all ash thermalises; setting it close to unity (with a low neutron fraction, e.g. p-B11) approaches the aneutronic DEC limit. The full driver draw $P_{\text{drv}}/\eta_{\text{pin}}$ still enters the recirculating load — unlike the inductive case below, the thermal-mode driver charge is not recovered.

4.3.3 Pulsed Inductive DEC Conversion

For inductive recovery concepts (e.g. pulsed FRCs with capacitor-bank-driven compression), the driver energy circulates in an electromagnetic loop: capacitor bank → compression coils → plasma → coils → capacitor bank. Charged-particle energy does PdV work on the expanding plasmoid, which is recovered inductively. The fraction of net charged-particle power that couples as PdV work is

$$f_{\text{pdv}} = 1 - \left(\frac{r_{\min}}{r_{\max}}\right)^{2(\gamma-1)} \tag{34}$$

with $\gamma = 5/3$ for an ideal monatomic plasma. The recovered and net DEC powers are then

$$P_{\text{recovered}} = \eta_{\text{dec}}\big(P_{\text{drv}} + f_{\text{pdv}}\, P_{\text{ch,net}}\big), \tag{35}$$

$$P_{\text{dee}} = P_{\text{recovered}} - P_{\text{drv}}, \tag{36}$$

where $P_{\text{ch,net}} = (1 - f_{\text{rad}})\, P_{\text{cp}}$ and η_{dec} is the round-trip efficiency of the inductive recovery circuit.

A key distinction from the steady-state balance is the recirculating load: because the capacitor bank energy goes out and comes back each cycle, only the charging *losses* $P_{\text{drv}}\,(1/\eta_{\text{pin}} - 1)$ are drawn from the grid, not the full $P_{\text{drv}}/\eta_{\text{pin}}$.

Neutrons, bremsstrahlung, and undirected charged particles still enter a thermal blanket. When $\eta_{\text{th}} > 0$ a conventional thermal cycle runs in parallel with the DEC, and the total gross electric is $P_{\text{et}} = P_{\text{dee}} + P_{\text{the}}$. When $\eta_{\text{th}} = 0$ the concept is purely inductive and all thermal loads (pumps, balance of plant (BOP)) are zeroed automatically.

For inductive DEC, the capacitor bank is costed on a \$/J stored-energy basis (Account 22.01.07, section 5.7): $C_{220107} = c_{\text{cap}} \times E_{\text{stored}}$, where c_{cap} is the all-in installed cost per joule. The NOAK default and its justification are discussed in section 5.7.

For the DEC inverse, the model takes the scientific gain Q_{sci} as input and derives the required stored energy per pulse:

$$E_{\text{stored}} = \frac{P_{\text{net}}}{f_{\text{rep}}\left(\eta_{\text{eff}}\left(1 + Q_{\text{sci}}\, f_{\text{cp}}\right) - 1\right)\eta_{\text{pin}}} \tag{37}$$

where η_{eff} combines the DEC and thermal conversion efficiencies. This ensures that higher DEC efficiency correctly reduces the required capacitor bank size rather than inflating it.

5 Cost Account Structure

The cost account structure maps the financial framework of section 3 onto specific plant systems and activities. Capital accounts CAS10–CAS60 sum to the total capital investment (CAPEX); annualized operating accounts CAS70 and CAS80 constitute OPEX; and CAS90 annualizes the CAPEX via the CRF. Table 4 shows the full hierarchy.

The *Disposition* column flags whether the account uses prior conventions unchanged (*Inherited*), keeps the prior backbone with additional sub-cases (*Extended*), or has been re-derived from procurement and first principles (*Replaced*). Subsections in this chapter present methodology only for accounts in the latter two categories.

The following subsections detail the methodology for each account that has undergone independent review and justification.

5.1 CAS10: Pre-Construction Costs

CAS10 covers land acquisition, site and plant permits, licensing, engineering studies, and pre-construction reports. It is typically less than 1% of total capital; individual sub-accounts do not warrant detailed parametric modeling.

5.1.1 Land and Land Rights (C11)

Modern compact fusion plant designs require 20–200 acres, far less than the legacy 1,000-acre assumption in earlier ARIES studies. A representative compact pilot-plant filing [23] plans 100 acres for 400 MWe, yielding a land intensity of 0.25 acres/MWe. At US industrial-zoned land costs of approximately \$10 000/acre (consistent with USDA farmland averages scaled for industrial zoning[1]), land cost for a 1 GWe plant is approximately \$2.5M, negligible in the context of multi-billion-dollar capital investments.

The compact siting is enabled by the absence of an exclusion zone: the NRC's 2023 decision to regulate D-T fusion under 10 CFR Part 30 (byproduct materials) rather than Part 50/52 (reactor licensing) eliminates the large-radius emergency planning zone required for fission plants [24]. Aneutronic fuels such as p-^{11}B likely fall outside NRC jurisdiction entirely.

5.1.2 Plant Licensing (C13)

Licensing timelines are fuel-dependent and affect first-of-a-kind (FOAK) builds by extending the total project time, which flows into the levelized annual cost calculation for CAS70 and CAS80. Table 5 summarizes the adopted values.

5.2 CAS21: Buildings and Structures

CAS21 covers all buildings and structures on the fusion plant site. Each building is priced individually per fuel type based on its physical scope and construction grade, not a blanket multiplier. All fusion plants are regulated under 10 CFR Part 30 (byproduct materials), not Part 50 (reactor license), which means no Seismic Category I requirements, no NQA-1 program for most buildings, and no fission-style containment structure.

5.2.1 Fuel-Dependent Scoping

Each building is costed for what it physically is. Key differentiators by fuel:

[1] USDA NASS, *Land Values 2024 Summary*, August 2024.

Table 4: Cost account structure overview. Accounts with dedicated methodology subsections are marked with †.

CAS	Description	Disposition	Method
10†	Pre-construction	Replaced	Land intensity scaling, fuel-dependent licensing
20	Total direct costs	–	Sum of CAS21–29
21†	Buildings & structures	Replaced	Per-building, per-fuel, industrial-grade (18 buildings)
22†	Reactor plant equipment	–	Component-level parametric (18 sub-accounts)
Per-module accounts (× n_{mod}):			
22.01.01†	First wall + blanket	Extended	Volume × thermal intensity × fuel × blanket-form structural complexity factor
22.01.02	Shield	Inherited	Volume × fuel-dependent neutron scale
22.01.03†	Coils (magnets)	Replaced	Conductor kAm × $/kAm × markup; $0 for IFE
22.01.04†	Heating (MFE) / driver (pulsed)	Extended	Per-MW heating; pulsed driver per-MJ (laser/accelerator) or per-MW (mechanical)
22.01.05	Primary structure	Inherited	Volume × power scale
22.01.06†	Vacuum system	Extended	Vessel volume scale + gas-load pumping ($S = Q/P_{\text{op}}$)
22.01.07†	Power supplies	Replaced	Power-scaled (MFE); $/J stored basis for all pulsed
22.01.08	Divertor / target factory	Inherited	Thermal-scaled (MFE) or power-scaled (IFE)
22.01.09	Direct energy converter	Replaced	Mirror/FRC override; circuit-derived markups for inductive DEC
22.01.10†	Remote handling & maintenance	Replaced	Fuel- and concept-dependent
22.01.11	Installation labor	Inherited	14% of reactor subtotal
22.01.12†	Isotope separation plant	Replaced	Zeroed; market purchase in CAS80
Plant-wide accounts:			
22.02	Main & secondary coolant	Inherited	Power-scaled
22.03	Auxiliary cooling + cryoplant	Inherited	Power-scaled + cryo load basis
22.04	Radioactive waste management	Inherited	Thermal-scaled
22.05	Fuel handling & storage	Inherited	Fuel-dependent, power-scaled (0.7)
22.06	Other reactor plant equip.	Inherited	Power-scaled (0.8)
22.07	Instrumentation & control	Inherited	Thermal-scaled (0.65)
23†	Turbine plant equipment	Replaced	Linear scaling on thermal electric P_{the} (NETL-calibrated)
24†	Electric plant equipment	Replaced	Linear scaling on gross electric (NETL-calibrated)
25†	Miscellaneous plant equip.	Replaced	Linear scaling on gross electric (NETL-calibrated)
26†	Heat rejection systems	Replaced	Linear scaling on total thermal P_{th} (NETL-calibrated)
27†	Special materials	Extended	Volume-based blanket-fill inventory: $V_{\text{blk}}\times$ vol-frac × density × $/kg (PbLi / Li / FLiBe / Be ceramic / ceramic only / none)
28†	Digital twin	Replaced	Fixed cost ($5M)
29†	Contingency on direct costs	Inherited	10% FOAK, 0% NOAK (Gen-IV EMWG)
30†	Indirect service costs	Replaced	Fraction of CAS20, construction-time scaled
40†	Owner's costs	Replaced	Fuel-dependent, staffing-based, power-law scaled
50†	Supplementary costs	Replaced	Fuel-dependent sub-account model (6 components)
60†	Interest during construction	Inherited	Uniform-spending IDC formula
70†	Annualized O&M + replacement	Replaced	Growing-annuity levelization
80†	Annualized fuel cost	Replaced	Fuel-specific consumables, burn-fraction corrected, levelized
90†	Annualized financial costs	Inherited	CRF × total capital

Table 5: Licensing timeline by fuel cycle.

Fuel	Licensing time (yr)	Rationale
D-T	2.0	Part 30 licensing, upper end of 1–2 yr range
D-D	1.5	Reduced tritium inventory
D-^{3}He	0.75	Minimal radioactivity
p-^{11}B	0.0	No NRC jurisdiction

Table 6: CAS21 buildings cost by fuel cycle (1 GWe reference).

Fuel	$/kW	M$ at 1 GWe	Construction grade
D-T	629	629	Enhanced industrial (tritium confinement, hot cell, rad-HVAC)
D-D	576	576	Enhanced industrial (reduced tritium/activation scope)
D-^{3}He	452	452	Light enhanced industrial
p-^{11}B	385	385	Industrial (no hot cell, no tritium, standard HVAC)

- **Hot cell:** D-T \$104M (full remote handling, 3–4 ft shielding, NQA-1 on tritium-wetted systems); D-D \$78M (3/4 scope, NQA-1 overhead similar); D-^{3}He \$23M (shielded inspection room only); p-^{11}B **\$0** (does not exist; nothing activates). The hot cell is physically integrated within the reactor building, following UK pilot-plant cost models, not a separate structure.
- **Reactor building:** D-T \$138M (biological shielding walls, tritium confinement barriers); p-^{11}B \$81M (standard heavy industrial crane hall, no shielding, personnel walk-in access).
- **Ventilation/HVAC:** D-T \$17M (rad-HVAC: HEPA, stack monitoring, emergency isolation); D-D \$15M (full rad-HVAC for tritium zones); p-^{11}B \$3.5M (standard industrial HVAC).
- **Security:** Tritium is not listed in 10 CFR Part 37 Category 1/2 table, so no armed response requirements apply for any fusion fuel. All fuels use standard industrial security (\$2–3.5M).
- **Site improvements:** D-T \$85M (larger footprint from hot cell, tritium processing); p-^{11}B \$69M (standard industrial site).

5.2.2 Scaling Basis

Different buildings scale with different physical drivers. The turbine building scales with thermal electric power P_{the} (steam cycle only, not DEC); the reactor building, hot cell, reactor auxiliaries, and ventilation/HVAC scale with fusion power P_{fus} (ventilation because its dominant load is confinement rad-ventilation of the reactor building, not office comfort HVAC); site improvements, control room, and security are approximately fixed; BOP electrical buildings scale with gross electric P_{et}. The administration building tracks staff and so scales sub-linearly, as $P^{0.5}$, consistent with the staffing-based accounts (CAS40, CAS70).

All power-scaling buildings are driven by *plant-total* power. When a plant is assembled from n_{mod} replicated modules, the power-scaling buildings are fed n_{mod} times the per-module power, the same total the balance-of-plant accounts (CAS23–26) use, so the buildings/site reach plant scale rather than being pinned to a single module; fixed buildings are charged once per site.

5.2.3 Comparison with Benchmarks

Table 7 situates the fuel-dependent CAS21 figures of table 6 against non-fusion plant types.

Table 7: CAS21 buildings $/kW benchmark comparison.

Plant type	Buildings $/kW	Grade
Gas combined-cycle (CCGT)	150–250	Industrial
Coal (NETL B12A)	175–300	Heavy industrial
Fission nuclear (Part 50)	800–1,300	Nuclear-grade

The fusion p-^{11}B figure of 384 \$/kW sits at roughly 1.7× CCGT, and the D-T figure of 626 \$/kW at roughly 3.0× CCGT. The values are derived from-scratch using industrial construction benchmarks [25, 26, 27], superseding ARIES/pyFECONS values.

5.3 Inherited Sub-Accounts

Sub-accounts CAS22.01.02 (shield), 22.01.05 (primary structure), 22.01.08 (divertor), and 22.01.11 (installation labor) inherit the methodology of the pyFECONS implementation [4] unchanged: hybrid volume + thermal-intensity scaling for component accounts and percentage-of-subtotal for installation labor. They are not re-derived here.

The divertor sub-account (CAS22.01.08) keeps this inherited scaling for steady-state MFE concepts that exhaust onto a material target, $C_{220108} = c_{\text{div}}\,(P_{\text{th}}/1000)^{0.5}$ with $c_{\text{div}} = 60$ M\$, but its value is concept-dependent. A levitated dipole has a closed-field-line topology and exhausts through the loss cone at the chamber openings, so it carries no manufactured divertor cassette and $C_{220108} = 0$. For inertial and magneto-inertial concepts the same sub-account instead funds the on-site target factory, built up from its three independent cost drivers,

$$C_{220108} = c_{\text{fix}} + c_{\text{line}}\,f_{\text{rep}} + c_{\text{flow}}\,(P_{\text{fus}}/1000)\,: \tag{38}$$

a fixed building and tritium-confinement shell (c_{fix}); precision production lines that scale with target throughput, i.e. the repetition rate f_{rep} (c_{line}); and material handling, cryogenics, and post-shot recovery that scale with mass-energy flow P_{fus} (c_{flow}). These are the only two independent factory variables: fixing the plant electric power and rep rate fixes the fusion power and hence the yield per shot ($P_{\text{fus}}/f_{\text{rep}}$ at fixed module count), so target size enters through P_{fus} alone and needs no separate term. This replaces an earlier flat \$244 M scaling on electric power—a fair proxy for the handling term, but blind to the throughput (rep-rate) term that is the axis the original basis was criticized for missing. The coefficients are a *per-concept* bottom-up build-up. A laser or heavy-ion capsule factory is a precision cleanroom whose tritium-handling back half (DT fill, ice-layering, contained assembly, injection) adds a dual-confinement premium and whose throughput is oversized by the inverse acceptance yield, totalling ∼725–780 M\$ at the design point and calibrated so its optimistic corner reproduces the LIFE [28] target factory (∼\$600 M, \$0.25/target). A MagLIF or Z-pinch load is instead a stamped-and-filled metal liner, a casting-shop process near ∼150 M\$—five times cheaper, and not the capsule cleanroom that the flat number phantom-applied to every concept. That flat figure, inherited from the Miles/LIFE factory total via pyFECONS [4], also double-counted the factory: the per-shot target cost (section 5.21) is sourced from fully-loaded studies that already amortize the plant, so the capital and per-shot accounts are disentangled here to count it once. Whether a concept carries a factory is gated on the per-shot target cost $c_{\text{target}} > 0$ and sized by c_{fac}: concepts that form their plasma or liner in situ—plasma-jet MIF, the liquid-liner magnetized-target loop, pulsed FRC, theta pinch, dense plasma focus, and sheared-flow / staged Z-pinch—set both to zero and carry $C_{220108} = 0$, with no cost override required.

Sub-account CAS22.01.01 (first wall + blanket) follows the pyFECONS volume × thermal-intensity × fuel-dependent unit-cost form, with the D-T reference unit cost anchored to a RAFM-steel blanket materials mass build-up (about 3500 t of fabricated nuclear-grade structure, breeder, and multiplier, NOAK-reduced and cross-checked against the 440-module ITER blanket/shield), extended with a blanket-form structural complexity factor $s(\text{form}) \in \{1.0, 1.3, 1.2, 0.0\}$ for liquid-metal, molten-salt, solid-breeder, and no-blanket architectures respectively:

$$C_{22.01.01} = c_{\text{blk}}(f) \cdot s(\text{form}) \cdot V_{\text{blk}} \cdot \left(\frac{P_{\text{th}}}{P_{\text{th,ref}}}\right)^{0.6} \tag{39}$$

The structural factor captures the cassette/canister fabrication premium for molten-salt designs (Hastelloy-N corrosion liner on FLiBe-wetted surfaces) and solid-breeder designs (pebble-bed canisters with separate breeder and multiplier zones), all relative to the PbLi-style flow-channel baseline. The fill material chemistry is costed separately under CAS27 (section 5.13).

The fuel-keyed unit cost $c_{\text{blk}}(f)$ is calibrated to a PbLi liquid-metal blanket (about 9.4 t/m^3, with a flow loop, MHD ducts, and online tritium extraction). A solid Li_2O ceramic breeder (used by the levitated dipole) has none of these, runs at about 2 t/m^3, and is roughly 3–5× cheaper per unit volume, so concepts with a Li_2O fill substitute a dedicated unit cost $c_{\text{Li}_2\text{O}} = 0.2$ M\$/m^3 in place of $c_{\text{blk}}(f)$.

Multi-Unit Labor Factor for $n_{\text{mod}} > 1$

Per-module installation labor (CAS22.01.11) is computed at the inherited 14% of reactor subtotal for the first unit at a site. For a multi-module site, the on-site labor for each subsequent unit is discounted by a fixed factor ϕ_{lab}:

$$C^{\text{site}}_{220111} = C^{\text{unit 1}}_{220111} \times \big(1 + (n_{\text{mod}} - 1)\,\phi_{\text{lab}}\big), \qquad \phi_{\text{lab}} = 0.92 \text{ (default)}. \tag{40}$$

Equipment costs (CAS22.01.01–22.01.10) are *not* discounted: each module is a separate manufactured copy and incurs full equipment cost. Plant-wide accounts (CAS22.02–22.07) already use total plant power $n_{\mathrm{mod}}\,P$ with sub-linear scaling exponents and capture the shared-infrastructure economy through that channel.

The factor ϕ_{lab} represents the documented twin/triplet unit co-location effect in fission EPC: re-use of construction crew, supervision, tooling, procedures, and engineering deliverables between adjacent units at a single site. The empirical anchors are roughly 20–30% labor reduction for Vogtle 3→4 (US AP1000), 10–15% across Korean APR-1400 batches (Shin-Kori, Barakah), and a smaller but positive effect on the Chinese AP1000 pairs (Sanmen, Haiyang). The European EPR pairs (Olkiluoto, Flamanville, Hinkley Point C) show no clear twin-unit effect in public data, with FOAK and regulatory churn dominating. The 8% default sits below the middle of the 5–15% empirical range.

This factor is deliberately *not* a Wright's-Law learning curve. Wright-style cost reductions apply to fleet-cumulative manufactured production (typically requiring fleet sizes in the hundreds to thousands), and the empirical fission literature on cross-program deployment finds no positive cross-program learning, with some analyses identifying *negative* learning at the program level for US and French nuclear [29, 30, 6]. Cross-fleet learning, if introduced in future work, belongs at a separate parameter representing cumulative units of a given concept built worldwide, not on n_{mod}, which represents units at a single site.

5.4 CAS22.01.03: Coils (Superconducting and Resistive)

CAS22.01.03 covers the magnet system: toroidal field (TF), central solenoid (CS), poloidal field (PF) coils, structural casing, insulation, quench protection, cryostat interfaces, and testing.

5.4.1 Conductor Scaling Law

The conductor quantity in kA-m follows from the magnet topology. For a toroidal system the on-axis field fixes the total ampere-turns through Ampère's law, $NI = 2\pi R_0 B/\mu_0$, while the conductor length per turn scales with the coil-bore radius r_{coil} (the winding encircles the plasma, blanket, shield, and vessel). The kA-m, and hence the cost, are therefore *bilinear* in the major radius R_0 and the coil bore r_{coil}:

$$C_{220103} = \frac{G \cdot B \cdot R_0 \cdot r_{\text{coil}}}{\mu_0 \times 1000} \times c_{\text{kAm}} \times M_{\text{concept}} \qquad \text{(toroidal).} \tag{41}$$

For a linear device (mirror, FRC, dipole, pulsed) the magnet is a true circular loop or solenoid, for which $B = \mu_0 NI/(2R)$ and the turn length is $2\pi R$, giving the textbook r^2 scaling $C_{220103} \propto G \cdot B \cdot r_{\text{coil}}^2/\mu_0$. Here G is a geometry factor (tokamak: $4\pi^2$; stellarator: $4\pi^2 \times f_{\text{path}}$, the $f_{\text{path}} = 2$ capturing the roughly doubled length of a non-planar 3D winding; mirror: $n_{\text{coils}} \cdot 4\pi$, the sum over independent solenoids), B is the field at the loop centre / on axis (not the peak field on the conductor, which is a factor of $\sim$2–4 higher for high-field superconductor but does not enter the ampere-metre quantity), c_{kAm} is the conductor cost per kA-metre, and M_{concept} is a manufacturing markup capturing winding, insulation, quench protection, structural casing, cryostat, and testing. Mirror coils are divided into two classes: $n_{\text{central}} = L/d_{\text{coil}}$ central-cell solenoids (continuous, spacing $d_{\text{coil}} = 5$ m by default) carrying ampere-metres at the coil design field b_{center}, plus $n_{\text{plug}} = 4$ high-field end-plug coils at the throat field $R_m B_{\min}$, where $B_{\min}$ is the plasma midplane field (b_{center} and $B_{\min}$ are independent inputs). The geometry factor $G = (n_{\text{central}} + n_{\text{plug}}) \cdot 4\pi$ accumulates over all independent solenoids.

The coil-bore radius r_{coil} is taken as the outer radius of the vacuum vessel from the device's radial build, since in a magnetic-confinement reactor the superconducting magnets sit outboard of the entire plasma–blanket–shield–vessel stack. The ARC point design places its TF coils beyond a 0.85 m inboard blanket/shield standoff [31], and ARIES-CS requires 1.5–2 m (1.79 m for a regular breeding module) between the plasma and the middle of the coil winding pack [32]. Deriving r_{coil} from the radial build, rather than treating it as a free parameter, ties the magnet cost to the same geometry that sets the blanket and shield volumes.

The mirror carries two distinct bores. The central-cell solenoids enclose the full radial build, so their bore is the vessel outer radius plus an assembly standoff, $r_{\text{coil}} = r_{\text{vessel,out}} + 0.10$ m, and they sit at the coil design field b_{center}. The end-plug coils sit at the throat, where the plasma necks down by flux conservation and no breeding blanket extends, so their bore is set by the throat plasma radius plus a throat standoff, $r_{\text{coil}} = a/\sqrt{R_m} + 0.30$ m, and they carry the throat field $R_m B_{\min}$. Small-bore high-field plug coils and large-bore lower-field central solenoids are the cost-relevant distinction unique to the mirror's two-class magnet set.

5.4.2 Conductor Cost Assumptions

The default conductor is REBCO high-temperature superconductor at \$50/kA-m, an aggressive NOAK target. Current market prices for REBCO tape are \$150–300/kA-m; the \$50/kA-m assumption reflects projected cost reductions from

Table 8: CAS22.01.03 manufacturing markups and path factors by confinement concept.

Concept	Markup	Path factor	Ref. coil cost	Rationale
Tokamak	3.09×	1.0	\$516M	TF+CS+PF complexity; calibrated at the SPARC-class reference (B
Stellarator	5.87×	2.0	\$2,130M	Non-planar 3D coils; 1.9× the tokamak markup, the NCSX modular-
Mirror	1.7266×	1.0	\$513M	Two-class: $n_{\text{central}} = L/d_{\text{coil}}$ central solenoids at b_{center} with bore r_{coil}
Pulsed FRC	1.5×	1.0	n/a	Theta-pinch formation coils
Theta pinch	1.5×	1.0	n/a	Compression coils
Orbitron	1.5×	1.0	n/a	Electrostatic confinement
Polywell	2.0×	1.0	n/a	Polyhedral magrid
Dipole	3.0×	1.0	n/a	Floating coil + lift coils (see text)
MIF, IFE, Z-pinch	\$0 (no plant-scale confinement magnets)			Seed coils opt-in

scaled-up manufacturing (analogous to the solar PV learning curve). For comparison, Nb_3Sn (the ITER conductor) is \$7/kA-m at maturity, and NbTi (LHC heritage) is similar.

This is a deliberate design choice: the model targets NOAK economics for a mature fusion industry, not FOAK procurement costs. For FOAK sensitivity analysis, the conductor cost may be overridden to \$150–300/kA-m, which roughly triples the coil account. Unlike B-11 enrichment (where the FOAK/NOAK gap reflects a missing industrial supply chain requiring \$10,000/kg falling to \$75/kg), the REBCO cost trajectory is a manufacturing learning curve for an existing product, making the aggressive target more defensible.

5.4.3 Resistive Copper Coils

The \$/kA-m model is correct where an expensive superconductor dominates cost. For low-field resistive copper coils (an FRC at near-unity beta runs external fields of order 0.1–1 T) the conductor is cheap per kA-m, so that path collapses to a near-zero, unphysical figure while the machine still has tonnes of copper to wind and support. When the conductor material is copper, the account is instead a *mass build-up* that re-prices the same ampere-metres by mass:

$$C_{220103} = \frac{1}{10^6}\left(m_{\text{Cu}}\, p_{\text{Cu}}\, k_{\text{Cu}} + m_{\text{st}}\, p_{\text{st}}\, k_{\text{st}}\right), \tag{42}$$

with $m_{\text{Cu}} = (\rho_{\text{Cu}}/J)$ (ampere-metres), $\rho_{\text{Cu}} = 8960$ kg/m^3, current density $J = 5$ A/mm^2, copper price $p_{\text{Cu}} = 11$ \$/kg, fabrication markup $k_{\text{Cu}} = 3.5$, steel support mass $m_{\text{st}} = 0.6\, m_{\text{Cu}}$, $p_{\text{st}} = 6$ \$/kg, and $k_{\text{st}} = 3.0$. The ampere-metres are the same $G\, B_{\max} R_{\text{coil}}^2/\mu_0$ quantity as the superconducting path, so no new geometry assumption is introduced. The branch is selected by conductor material, so it applies to every copper concept (steady and pulsed FRC, theta pinch, polywell, orbitron) and leaves superconducting concepts on the \$/kA-m path. A steady FRC at $B = 0.5$ T, $R = 1.85$ m, $n_{\text{coils}} = 4$ carries about 123 t of copper and 74 t of steel, giving $C_{220103} \approx$ \$6M, versus about \$0.1M from the \$/kA-m path at the same field.

5.4.4 Levitated Dipole Floating Coil

The levitated dipole replaces the external magnet set with a single superconducting floating coil immersed in the plasma, so its CAS22.01.03 follows a dedicated build-up rather than eq. (41). The floating coil's conductor quantity is the single-ring value $4\pi\, B_{\text{center}}\, R_{\text{coil}}^2/\mu_0$ (geometry factor $G = 4\pi$ as for one mirror solenoid), priced on the same \$/kA-m conductor basis but carrying a higher markup ($M_{\text{lev}} = 8$) for the float, no-access engineering: a persistent-current joint, an integral cryostat, and inductive charging with no demountable leads. The external stationary lift coils that hold the floating coil against gravity are taken as a fixed fraction (default 0.10) of the floating-coil cost, and an integral cryostat term covering neon-slush cooling, the flux pump, and levitation control is added as a flat \$100M:

$$C_{220103} = M_{\text{lev}}\, c_{\text{float}} + f_{\text{lift}}\, M_{\text{lev}}\, c_{\text{float}} + c_{\text{cryo}}, \tag{43}$$

with $f_{\text{lift}} = 0.10$ and $c_{\text{cryo}} = 100$ M\$. The reference OpenStar Reactor A point uses $B_{\text{center}} = 6.26$ T and $R_{\text{coil}} = 5.3$ m.

5.5 CAS22.01.04: Supplementary Heating / Primary Driver

CAS22.01.04 covers different hardware depending on the confinement family.

5.5.1 Steady-State MFE

For tokamaks, stellarators, and mirrors, the account covers vendor-purchased auxiliary heating systems. Cost scales linearly with installed power:

$$C_{220104} = \sum_i c_i\, P_i \quad [\text{M\$}] \tag{44}$$

where $c_{\text{NBI}} = 7.06$, $c_{\text{ICRF}} = 4.15$, $c_{\text{ECRH}} = 5.00$, $c_{\text{LHCD}} = 4.00$ (all M$/MW, 2023$). The per-MW costs are calibrated to ITER procurement contracts, adjusted from FOAK to NOAK.

The split P_i is priced per MW of *injected* heating power, the same P_{in} the power balance uses ($q_{\text{sci}} = P_{\text{fus}}/P_{\text{in}}$); the wall-plug draw $P_{\text{in}}/\eta_{\text{pin}}$ enters the recirculating power, not this capital line. To keep the costed power identical to the power-balance power, the split is the heating *mix* and is normalized so $\sum_i P_i = P_{\text{in}}$; overriding P_{in} rescales the split, so the costed heating always tracks the physics. A fully zero split (electrostatic concepts whose input power is not NBI/RF heating, e.g. orbitron, polywell) stays zero, leaving their heating capital at $0. Pricing the supply-dominated NBI capital on wall-plug power $P_{\text{in}}/\eta_{\text{pin}}$, which would auto-elevate low-coupling concepts such as the FRC, is a deliberate refinement not applied here; the driver's recirculating burden is already represented through η_{pin}.

5.5.2 Pulsed Concepts

For pulsed concepts, the account covers the primary driver capital, namely the hardware that compresses or ignites the target. This is the pulsed analog of the magnet system (CAS22.01.03): the mechanism that provides confinement. The cost basis depends on the driver type.

Lasers, accelerators, and electromagnetic guns are costed per joule of pulse energy. Their capital is set by the per-pulse energy (laser aperture, amplifier and pump-diode count, accelerator beam energy and storage-ring charge, coaxial-gun size and peak current) and not by how often the driver fires. Repetition rate adds cooling and consumable replacement, which are sub-dominant capital and operating costs rather than a multiplier on the driver hardware itself. Costing on average power $E_{\text{drv}} f_{\text{rep}}$ would instead price one physical laser across the 0.1 to 15 Hz range concepts run at as though it were a hundred different machines. Hence

$$C_{220104} = c^{E}_{\text{drv}} \times E_{\text{drv}} \quad [\text{M\$}] \tag{45}$$

with c^{E}_{drv} in M$/MJ. Pneumatic and mechanical injectors (mag-target) instead accelerate target mass on every shot, so throughput (average power) drives their hardware count, handling, and recirculation plant, and they retain an average-power basis with $P_{\text{drv}} = E_{\text{drv}}\, f_{\text{rep}}$:

$$C_{220104} = c^{P}_{\text{drv}} \times P_{\text{drv}} \quad [\text{M\$}] \tag{46}$$

The heavy-ion per-joule coefficient reproduces the underlying $/W NOAK projection at its reference repetition rate ($12/W at 5 Hz gives 60 M$/MJ) and then holds constant away from that reference. The laser coefficient of 205 M$/MJ is set from published diode-pumped solid-state laser (DPSSL) NOAK estimates of $210–700/J: the CAS22.01.04 optics and diode array carry about $205/J, and the CAS22.01.07 capacitor bank that fires the diodes adds about $5/J, so the turnkey driver lands at the aggressive end (about $210/J) of the published NOAK range, consistent with a diode roadmap toward $0.007/W. The plasma-jet gun and the sheared-flow Z-pinch's coaxial gun follow the same per-pulse logic; the single sheared-flow gun assembly is cheaper per joule (1.5 M$/MJ) than the plasma-gun array (4 M$/MJ). The laser coefficient branches on driver architecture, selected by `laser_driver_type`: diode-pumped solid-state (DPSSL, the default) at 205 M$/MJ; KrF excimer at 40 M$/MJ, leaning to the large-aperture-optics projection for e-beam-pumped excimer (engineering range 20–200 M$/MJ); and flashlamp-pumped Nd:glass (NIF-class) at 1000 M$/MJ, set from the NIF $3.5–4.2 B / 1.1–1.9 MJ-UV facility cost.

Driver scheduled replacement. A rep-rated laser driver's replaceable subsystems wear at shot lifetimes spanning sub-annual to multi-decade, so each is charged as the level annual cost of replacing it every $t = N_{\text{life}}/\dot{N}_{\text{shot}}$ years over the plant life, summed by the same geometric replacement model used for the core, direct-energy-converter grid, and capacitor bank. The first set is capital (already in CAS22.01.04); only replacements beyond it are charged. For DPSSL the pump diodes carry a NOAK shot life ($\sim 10^{10}$) that exceeds a 30-year plant at 10 Hz, so they are effectively a capital item and the operating burden is dominated by the final optics; for flashlamp-pumped Nd:glass the lamp shot life ($\sim 10^{4}$) is sub-annual and the replacement term is prohibitive, which is the standard reason NIF-class drivers are not viable for rep-rated energy.

For MagLIF the main driver is the electrical Z-pinch, costed in CAS22.01.07; the only CAS22.01.04 contribution is an optional laser preheat, costed per joule of preheat pulse energy $c_{\text{pre}} E_{\text{pre}}$ on the same diode-pumped solid-state laser

Table 9: CAS22.01.04 driver cost coefficients by pulsed concept.

Concept	Basis	Coeff.	Hardware
Laser IFE	M$/MJ	205	Diode-pumped solid-state laser
Heavy ion	M$/MJ	60	RF linac + storage rings
Plasma jet	M$/MJ	4	Plasma gun array
Staged Z-pinch	M$/MJ	1.5	Coaxial gun + gas injection
MagLIF	M$/MJ	205	Laser preheat (DPSSL); main driver in CAS22.01.07
Mag. target	M$/MW	3	Pneumatic pistons, liquid metal loop
Z-pinch, DPF	n/a	0	Driver is electrical (CAS22.01.07)
Pulsed FRC, Theta pinch	n/a	0	Driver is coils (CAS22.01.03)

basis as the IFE driver. Magnetized-compression concepts that omit the preheat laser set $E_{\text{pre}} = 0$ and incur no preheat cost, while the coefficient remains available for those that include one.

Concepts whose driver is purely electrical (pulsed power / capacitor bank) have their driver capital captured entirely in CAS22.01.07 on a $/J stored basis, avoiding double-counting. Similarly, concepts whose confinement is provided by coils (pulsed FRC, theta pinch) have that hardware in CAS22.01.03.

5.6 CAS22.01.06: Vacuum System

CAS22.01.06 is a sum of two parts with different cost drivers:

$$C_{220106} = C_{\text{vessel}} + C_{\text{pump}}. \tag{47}$$

The *vessel shell* (welded stainless chamber, ports, gauges, leak detection) is volume-based, scaling with reactor size:

$$C_{\text{vessel}} = c_{\text{ves}}\, V_{\text{ves}} \left(\frac{P_{\text{et}}}{1100}\right)^{0.6}, \qquad c_{\text{ves}} = 0.72 \text{ M\$/m}^3. \tag{48}$$

The *pumping* system is sized by gas throughput, not vessel volume. A pump of speed S removes a throughput $Q = SP$, so the installed speed required to hold an operating (plenum) pressure P_{op} is $S_{\text{req}} = Q_{\text{gas}}/P_{\text{op}}$, costed at $c_{\text{pump}} = 0.015$ M$ per (m^3/s) (commodity cryopump procurement, $15/(L/s)). The gas load has two sources; wall outgassing scales with surface area but is negligible for a baked UHV system and is omitted:

$$Q_{\text{NBI}} = g\,\frac{P_{\text{NBI}}}{E_b}\,k_B T, \tag{49}$$

$$Q_{\text{fuel}} = n_{\text{ion}}\,\frac{1-f_b}{f_b}\,\frac{P_{\text{fus}}}{E_{\text{fus}}}\,k_B T, \tag{50}$$

with $T = 300$ K, gas amplification $g = 1$ (gas particles pumped per beam particle, calibrated to the C-2W machine's roughly 2000 m^3/s divertor pumping), reference NBI energy $E_b = 120$ keV (the NBI gas load scales as $1/E_b$), $n_{\text{ion}} = 2$ ions consumed per reaction, and E_{fus} the fuel-dependent energy per reaction. Then $C_{\text{pump}} = c_{\text{pump}}(Q_{\text{NBI}} + Q_{\text{fuel}})/P_{\text{op}}$.

The operating pressure P_{op} is the dominant, concept-specific driver, set per concept: a few Pa for tokamak and stellarator divertor plenums (high-recycling regions designed to pump at high pressure), versus 0.02–0.05 Pa for open-field-line linear devices (mirror, FRC) that must hold a low neutral pressure to limit charge-exchange losses. The same throughput therefore costs an FRC roughly $60\times$ more to pump than a tokamak. The fueling term dominates over the NBI term at reactor beam energies; because aneutronic p-B^{11} releases about half the energy per reaction of D-T, it runs about twice the reaction rate and pumps about twice the fuel throughput at equal fusion power. The pumping model applies uniformly to every concept, so these differences fall out of the inputs rather than per-concept hand-tuning.

Reporting convention. Where a single account aggregates components with distinct cost bases, the model emits informational sub-lines keyed `<CODE>_<component>` (here `C220106_vessel` and `C220106_pump`) alongside the canonical total. These sub-lines are excluded from all aggregation and exist only for visibility and sensitivity tracing. CAS22.01.06 is currently the sole account that uses them.

5.7 CAS22.01.07: Power Supplies

CAS22.01.07 covers the electrical power supply systems that drive the fusion core: magnet power supplies, pulsed power drivers, and associated switchgear. The account has two modes.

5.7.1 Steady-State MFE

For tokamaks, stellarators, and mirrors, the cost scales with gross electric output:

$$C_{220107} = 80 \times (P_{\text{et}}/1000)^{0.7} \quad [\text{M\$}] \tag{51}$$

The \$80M reference at 1 GWe originates from the ARIES-CS design study (\$70.6M in 2004\$ for a 1.3 GWe stellarator, approximately \$78/kW in 2024\$). Cross-checks against GW-scale HVDC converter stations (\$100–250/kW installed)[2] and utility-scale solar inverters (\$30–50/kW [27]) bracket the figure: fusion magnet supplies operate at lower voltage and higher current than HVDC, which is the cheaper regime for power electronics. No direct vendor quote exists at fusion-relevant parameters (1–10 kV, 10–100 kA, 100+ MW). Uncertainty: $\pm 20\%$.

5.7.2 Pulsed Concepts

For all pulsed concepts, the cost scales with stored energy per pulse on a \$/J basis:

$$C_{220107} = c_{\text{cap}} \times E_{\text{stored}} \quad [\text{M\$}] \tag{52}$$

where $c_{\text{cap}} = 0.50$ \$/J is the NOAK all-in installed cost (capacitors, switches, charging supplies, buswork). This is an aggressive target: present lab-scale pricing is \$20–50/J, a 40–100× gap. The CATF IWG extension [5] recommends \$1.5–4/J for NOAK. The \$0.50/J default assumes advanced-dielectric capacitors manufactured at high volume, a commercial viability requirement, not a current capability.

Capacitor banks require scheduled replacement (CAS72). At $f_{\text{rep}} = 1$ Hz and 85% availability, the bank executes 27 million cycles per year. At the NOAK baseline lifetime of 10^8 shots, replacement occurs every 3.7 years. The annualized replacement cost is computed as a present-value sum of discrete replacements over the plant lifetime, discounted at the project interest rate.

5.7.3 Source Provenance

The steady-state MFE figure draws partly on the ARIES-CS [32] power-supply scope, which is a systems study rather than a procurement reference. The capacitor-bank \$/J basis used for pulsed concepts is procurement-grounded (commercial high-voltage capacitor pricing). Substituting a procurement reference for the steady-state MFE figure is left to follow-on work; a vendor quote may be applied via `cost_overrides={"C220107": <value>}` until that substitution lands.

5.8 CAS22.01.09: Direct Energy Converter (electrostatic)

CAS22.01.09 covers add-on direct energy converters for linear devices (mirrors and steady-state FRCs) where charged-particle exhaust escapes along open field lines and can be decelerated to recover kinetic energy as electricity. Inductive DEC for pulsed-FRC architectures is captured in CAS22.01.07 via Q_{sci}-derived markups on the driver bank (section 5.7); the present subsection covers the electrostatic venetian-blind variant for steady-state mirrors.

The cost scales with DEC electric output:

$$C_{220109} = c_{\text{dec}} \times (P_{\text{dee}}/P_{\text{dee,ref}})^{0.7} \quad [\text{M\$}] \tag{53}$$

where $c_{\text{dec}} = 125$ M\$ is the NOAK all-in 2024 USD reference cost at $P_{\text{dee,ref}} = 400$ MWe DEC electric output, and the 0.7 exponent reflects sub-linear scaling of vacuum/tank infrastructure with throughput while grid modules and power conditioning scale roughly linearly.

The reference figure is a bottom-up build-up of the DEC-specific add-on hardware (excluding shared mirror infrastructure already costed under CAS22.01.06 and the expander):

- Tank: \$3–15 M, central \$7 M. Industrial-grade fabricated 304L stainless (\$10–15/kg cleaned and ASME Section VIII Div 1 certified for 10^{-5} to 10^{-6} Torr service). Sizing for atmospheric collapse load on a 10–15 m diameter, 20–40 m long shell with ring stiffeners every 1–2 m gives 300–500 tonnes (low end) to 600–800 tonnes (conservative). Same dimensional class as commercial refinery hydrocracker reactors [34] (4–5 m × 25–30 m, 1000–1500 tonnes, ASME VIII Div 2 at 200 bar internal) which run \$26–30 M; the VB tank holds only vacuum (1 atm external collapse load), so it sits well below that price. ITER cryostat (in the range \$30–40/kg, nuclear-grade, double-walled) is overspec on certification.

[2] HVDC converter station vendor literature, Siemens Energy and Hitachi ABB Power Grids, 2022–2024.

- HV power conditioning (DC-AC chain): \$35–55 M, central \$45 M. The DEC outputs DC at multi-stage retarding potentials (30–100 kV), which a converter chain returns to grid AC. MISO's 2024 MTEP24 cost guide [35] reports VSC HVDC valve hall capital at \$167/kW for ± 250 kV / 500 MW and \$178/kW for ± 400 kV / 1.5 GW; the DEC needs the IGBT/DC-conversion portion only (no duplicate AC interconnect), placing it at roughly 60–70% of full HVDC station cost, \$100–120/kW.
- Cryopumps: \$10–25 M, central \$15 M. 10^6–10^7 L/s pumping speed. ITER cryopumps procurement [36] (8 torus and cryostat cryopumps, EUR 21 M F4E investment, manufactured by Research Instruments / Alsymex) and industrial cryopump pricing[3] (\$200–500k per 20,000 L/s) bracket the figure.
- Heat collection panels: \$10–20 M, central \$15 M. Hypervapotron-cooled tungsten panels for the 30–90 MW heat load deposited on grids and collectors (5–15% of charged-particle power for geometrically transparent grids). Design analog: the ITER NBI residual ion dump and calorimeter [37], which handle 5–13 MW/m^2 heat flux on hypervapotron-cooled Cu-CrZr / W elements at the 16 MW beam-line scale. Public procurement values for the ITER beam-dump components are not disclosed; the dollar figure here is an order-of-magnitude estimate scaled by area and heat flux. ITER divertor pricing (\$60M/GW$_{\text{th}}$) is overspec, since VB grids are not exposed to direct fusion plasma or 14 MeV neutron flux.
- Grid and collector modules: \$10–20 M, central \$15 M. Tungsten ribbon grids, alumina HV insulators, stainless mounting frames. Geometry-driven; this line uses Hoffman [38] (\$4k/m^2 in 1975 USD, \$23k/m^2 in 2024 USD via CPI) at 300–1000 m^2 collector area. Cross-checked against the TMX-integrated 48% demonstration [39].
- Misc (HV bushings, vacuum gate valve, controls, instrumentation): \$10–17 M, central \$12 M.

The hardware subtotal is \$78–152 M, central \$109 M. Adding 15% installation (NOAK) gives \$90–175 M total, central \$125 M, which is the value used for c_{dec}. Uncertainty: $\pm 35\%$, dominated by the HV power conditioning scale-up factor and the plant-scale tank fabrication unit cost.

5.9 CAS22.01.10: Remote Handling & Maintenance Equipment

CAS22.01.10 covers the robotic and remotely-operated equipment required for in-vessel maintenance: manipulators, transfer casks, in-pipe welding/cutting tools, and rad-hardened actuators. The building that houses this equipment (hot cell) is costed separately in CAS21.

The cost model is:

$$C_{220110} = C_{\text{RH,base}}(f) \times S_{\text{concept}} \times \left(\frac{P_{\text{et}}}{1000}\right)^{0.5} \tag{54}$$

where $C_{\text{RH,base}}(f)$ is a fuel-dependent base cost (M\$ at 1 GWe tokamak reference) and S_{concept} is a geometry-dependent scaling factor.

5.9.1 Fuel Dependence

Neutron activation determines whether remote handling is needed and how heavily equipment must be rad-hardened. Table 10 lists the base costs.

Table 10: Remote handling base costs by fuel cycle (1 GWe tokamak reference).

Fuel	Base (M\$)	Rationale
D-T	150	Full RH suite, rad-hardened to 14.1 MeV neutrons
D-D	100	Same scope, reduced rad-hardening (2.45 MeV)
D-^{3}He	30	Lighter duty, activation still exceeds occupational limits
p-^{11}B	20	Conventional maintenance equipment (no rad-hardening)

5.9.2 Concept Dependence

Toroidal vessels (tokamak, stellarator) require articulated multi-axis manipulators to reach components through narrow ports, while linear geometries (mirror, FRC) permit simpler end-access tooling. Toroidal concepts use $S_{\text{concept}} = 1.0$; linear concepts use $S_{\text{concept}} = 0.55$.

[3] Edwards/Leybold STP-iXR class turbomolecular and cryogenic pump product datasheets, list pricing 2023.

5.10 CAS22.01.12: Isotope Separation

CAS22.01.12 is set to zero for all fuel cycles. All isotope procurement is modeled as market purchase in CAS80 at enriched unit prices.

The rationale is that CAS22.01.12 (on-site separation plant capital) and CAS80 (enriched market prices) are mutually exclusive models of the same cost. If CAS80 pays $2,175/kg for deuterium, that price already includes the supplier's extraction and enrichment costs. Simultaneously budgeting a $300M on-site heavy-water plant charges for separation twice. The correct model is one or the other:

- **Market purchase:** CAS80 at enriched $/kg, CAS22.01.12 $= 0$.
- **On-site plant:** CAS22.01.12 capital cost, CAS80 at raw feedstock prices.

Market purchase is adopted because on-site separation is not justified at fusion-scale consumption rates. A 1 GWe D-T plant consumes 73–194 kg/yr of deuterium against a global enriched supply exceeding 80,000 kg/yr from heavy-water plants. Tritium is bred on-site in the blanket and processed by the fuel handling system (CAS22.05); it is not a market purchase. Lithium-6 enrichment capacity (1–2 t/yr globally) could be stressed by a fleet of D-T reactors, but a centralized enrichment facility shared across plants is a fleet-level investment, not a per-plant CAS22 capital item.

Table 11 lists the retained CAS80 unit costs.

Table 11: Fuel isotope unit costs (CAS80, market purchase).

Isotope	$/kg	Basis
Deuterium	2,175	STARFIRE (1980), inflation-adjusted
Li-6 (enriched)	1,000	90% enriched Li-6
He-3	2,000,000	Scarcity pricing
Protium	5	Commodity H_2
B-11 (enriched)	10,000	FOAK estimate (B-10 tails)

5.11 CAS22.02–.07: Plant-Wide Reactor Systems

The plant-wide reactor systems CAS22.02 (main and secondary coolant), CAS22.03 (auxiliary cooling and cryoplant), CAS22.04 (radioactive waste management), CAS22.05 (fuel handling and storage), CAS22.06 (other reactor plant equipment), and CAS22.07 (instrumentation and control) inherit pyFECONS power-scaling laws with exponents in the range 0.65–0.85 and pyFECONS reference-plant unit costs. CAS22.05 is fuel-dependent (D-T tritium accounting versus aneutronic fuel handling); the others are concept- and fuel-agnostic.

5.12 CAS23–26: Balance of Plant Equipment

CAS23–26 cover conventional power-conversion and electrical infrastructure. These accounts are **fuel-independent**: the choice of fuel affects the reactor island (CAS22) but not the turbine hall, switchyard, or cooling towers.

The model supports three thermal cycles, selected by the user: steam Rankine, supercritical CO_2 (sCO_2) Brayton, and combined cycle (gas topping plus steam bottoming). The cycle determines the thermal conversion efficiency η_{th} and the per-MW coefficients for CAS23 (turbine plant) and CAS26 (heat rejection). CAS24 (electrical equipment) and CAS25 (miscellaneous plant) are cycle-independent.

CAS23 (turbine plant) scales with thermal electric power $P_{\text{the}} = \eta_{\text{th}} P_{\text{th}}$, and CAS26 (heat rejection) scales with total thermal power P_{th}. This ensures both accounts auto-zero for pure direct energy conversion plants ($\eta_{\text{th}} = 0$) and scale correctly for hybrid DEC-plus-thermal configurations. CAS24 and CAS25 scale with gross electric P_{et}:

$$C_{23} = n_{\text{mod}}\, P_{\text{the}}\, c_{23}, \quad C_{26} = n_{\text{mod}}\, P_{\text{th}}\, c_{26}, \quad C_{24,25} = n_{\text{mod}}\, P_{\text{et}}\, c_{2x} \tag{55}$$

where c_{2x} is a per-MW coefficient (M$/MW, 2024$). Table 12 lists the adopted values for each cycle.

The Rankine coefficients derive from the ARIES cost-account tradition [40], calibrated against the NETL *Cost and Performance Baseline for Fossil Energy Plants* [26]. Pre-inflation (2019$) values are escalated by a CPI factor of 1.22.

The sCO_2 Brayton efficiency of $\eta_{\text{th}} = 0.47$ reflects the DOE/Sandia target of 45–50% for recompression Brayton cycles at nuclear heat source temperatures (500–700°C) [41, 42]. The CAS23 coefficient is 22% below Rankine: sCO_2 turbomachinery is 85% smaller by volume, but recuperators add cost, netting 20% overall reduction [43].

Table 12: Balance of plant per-MW coefficients by thermal cycle (2024$).

CAS	Scope	Rankine	sCO_2	Combined
23	Turbine plant equipment	0.198	0.155	0.235
24	Switchyard, transformers, cabling	0.084	0.084	0.084
25	Fire protection, compressed air, HVAC	0.051	0.051	0.051
26	Cooling towers, circulating water	0.034	0.022	0.018
	η_{th}	0.40	0.47	0.53

The combined cycle efficiency of $\eta_{\text{th}} = 0.53$ is derived from NGCC performance (>60%, National Energy Technology Laboratory 26), adjusted downward for fusion heat source temperatures (600°C vs. 1500°C gas turbine inlet). The CAS23 coefficient is 19% above Rankine due to dual turbomachinery sets (topping plus bottoming) and a heat recovery steam generator (HRSG).

Cross-checks against the NREL ATB 2024 nuclear cost breakdown [27] (3.9% of total capital for energy conversion, 6.3% for electrical equipment) confirm that the adopted Rankine values are conservative, appropriate for NOAK fusion plants using conventional, non-nuclear-grade BOP equipment.

5.13 CAS27: Special Materials

CAS27 covers the initial inventory of non-fuel reactor materials: breeding blanket fill, neutron multiplier, and other special inventory. CAS22.01.01 covers the blanket *structure*; CAS27 covers the *material that fills it*. A material inventory is set by blanket volume, not net electric power, so CAS27 is a volume-based mass build-up keyed on the blanket fill:

$$C_{27} = V_{\text{blk}} \cdot \nu(\text{fill}) \cdot \rho(\text{fill}) \cdot c(\text{fill}) \tag{56}$$

where V_{blk} is the first-wall + blanket + reflector volume, ν the fraction of that region occupied by the costed material (liquid fill fraction; or breeder/multiplier-zone $\times$ pebble packing ≈ 0.25 for solid pebble beds), ρ its density, and c its unit price.

The blanket configuration is specified as an orthogonal pair: a `BlanketForm` (structural architecture, driving $s(\text{form})$ in CAS22.01.01) and a `BlanketFill` (bulk material, driving the inventory here). Compatibility between form and fill is enforced at validation time. Fuel-appropriateness enters through the chosen fill: aneutronic / non-breeding concepts use `none` and recover zero. Table 13 lists the per-fill inputs.

Table 13: CAS27 blanket-fill inventory inputs (volume-based build-up).

Fill	ρ (kg/m^3)	ν	c ($/kg)	Basis
`pbli`	9400	0.50	5	Pb-17Li (99.3% Pb) + enriched Li-6 premium
`li`	490	0.80	200	liquid Li, Li-6 enrichment ($80–1000/kg)
`flibe`	1940	0.80	150	2LiF-BeF_2, Be-dominated (NOAK)
`be_ceramic`	1850	0.25	700	HCPB Be multiplier; Li-ceramic folded in
`ceramic_only`	2400	0.25	150	Li-ceramic breeder pebbles
`li2o`	2013	0.25	150	Li_2O ceramic
`none`	–	–	–	Aneutronic, no breeder

Scaling by the modelled blanket volume rather than a power-law keeps CAS27 consistent with the blanket structure (CAS22.01.01) and the other volume-based reactor-component accounts, and it costs both a compact high-power-density design (e.g. an ARC-class FLiBe immersion blanket, $\sim$100M) and a large thin-shell design (e.g. a levitated dipole's Li_2O blanket) by their actual material volume.

The HCPB (`be_ceramic`) inventory is dominated by global beryllium supply: world production is 300 tonnes/yr, so a single DEMO-scale plant consumes the entire annual supply, making the beryllium cost the largest known discriminator in the EUROfusion blanket selection process.

Aneutronic fuels (D-He3, p-B11) typically use `none/none` (no breeding blanket), recovering minimal special-material inventory.

5.14 CAS28: Digital Twin

CAS28 covers a plant-wide digital twin for operator training, predictive maintenance, and operational optimization. A fixed cost of $5M is adopted, independent of plant size, fuel type, or number of modules: the digital twin is software-dominated, and its complexity scales with the number of modeled subsystems rather than plant capacity.

The $5M figure is anchored to disclosed advanced-reactor digital-twin budgets. Under the DOE ARPA-E GEMINA program [44], the University of Michigan received $5.2M to develop a scalable reactor digital twin (first validated on a campus molten-salt loop, then applied to the Kairos Power design), alongside a complementary $2.2M Argonne award for O&M automation. A single plant-wide reactor twin therefore lands at the adopted $5M, consistent with industrial digital twin deployments for large thermal plants ($2–10M; Siemens MindSphere, GE Predix, AVEVA). The figure may be varied via `cost_overrides={"CAS28": <value>}`.

5.15 CAS29: Contingency on Direct Costs

CAS29 is a percentage-based contingency allowance on the sum of CAS21–28:

$$C_{29} = f_{\text{cont}} \times \sum_{i=21}^{28} C_i \tag{57}$$

where $f_{\text{cont}} = 10\%$ for FOAK and 0% for NOAK, following the Gen-IV EMWG convention [3]. The 10% FOAK contingency is conservative relative to the EMWG recommendation of 15–20% for advanced reactors; individual CAS22 sub-accounts already include conservative fabrication markups. The 0% NOAK contingency reflects the assumption of a mature, standardised design with resolved construction risk.

5.16 CAS30: Capitalized Indirect Service Costs

CAS30 covers construction support services not attributable to specific plant systems: field indirect costs (equipment rental, temporary structures, consumables), construction supervision, and offsite design services [3].

5.16.1 Literature Review

Three methodologies exist in the fusion costing literature:

1. **Schulte (1978):** A fixed 35% of total direct cost (TDC), split as 15% construction supervision, 15% design services, 5% field indirect costs [1].
2. **Miller/ARIES LSA fractions:** A Level of Safety Assurance (LSA) dependent fraction of TDC, ranging from 21.7% at LSA=1 (inherently safe, minimal nuclear-grade requirements) to 29.0% at LSA=4 (full nuclear-grade construction) [2, 45, 46]. The LSA concept [45] captures the cost reduction from eliminating nuclear-grade construction requirements where the physics permits.
3. **pyFECONS bottom-up:** A power-scaling formula calibrated from a staffing model for a 150 MWe reference plant [4]. This yields 5.8% of TDC at 1 GWe, reflecting the labor component; non-labor indirect costs (equipment rental, temporary structures, consumables) that dominate CAS31 scope are not included in the staffing model.

No real fusion construction data exists to calibrate CAS30. Private fusion companies do not publish cost breakdowns at this granularity; ITER is a one-of-a-kind research facility; and UK STEP and EU-DEMO cost studies use ARIES-lineage models. For context, efficient series-built fission plants (Korean APR1400, Chinese Hualong One) achieve indirect costs of 12–20% of TDC, while troubled FOAK projects (US Vogtle 3&4) exceed 35%.

5.16.2 Adopted Model

A simpler aggregate model is adopted:

$$\text{CAS30} = f_{\text{indirect}} \times \text{CAS20} \times \frac{T_c}{T_{\text{ref}}} \tag{58}$$

where $f_{\text{indirect}} = 0.20$ (default), T_c is the construction time, and $T_{\text{ref}} = 6$ yr is a reference duration. The construction-time scaling captures the primary driver that varies between scenarios.

The default of 20% is chosen as:

- Below Schulte's 35% (1978 fission baseline, too high for NOAK fusion),
- Below Miller's LSA=2 value of 23.2% (ARIES tradition, likely somewhat high for NOAK),
- Above pyFECONS's 5.8% (labor-only staffing model),
- Consistent with efficient NOAK nuclear construction internationally (15–20% of TDC).

Sub-account precision (C31/C32/C35 splits) is not empirically distinguishable for fusion and is not modeled.

5.17 CAS40: Capitalized Owner's Costs

CAS40 covers one-time costs incurred by the plant owner during construction to build an operating organization before commercial operation: staff recruitment and training (C41), staff housing/relocation (C42), and salary-related costs during the pre-operational period (C43) [3, 47].

5.17.1 Relationship to CAS70

CAS40 and CAS70 share the same underlying staff pool but cover different time periods. CAS40 is the capital cost of *building* the operations team (pre-COD); CAS70 is the annual cost of *running* it (post-COD). Both are derived from the same fuel-specific staffing analysis (section 5.20), applied through different cost models. There is no double-counting.

5.17.2 Methodology

The INL first-principles SFR cost study [47] estimates CAS40 sub-accounts as:

- C41: 1.5 years of salary for 110% of staff (IAEA-recommended overhire for training attrition), plus a 25% recruiting fee.
- C42: $40k relocation cost for 50% of staff.
- C43: Benefits at 57–59% of salary during the training period.

Applying this methodology to the fuel-specific staffing estimates yields total pre-operational costs that scale as roughly 3.2× annual labor-plus-benefits.

5.17.3 Adopted Model

$$\text{CAS40} = C_{\text{owner}}(f) \times \left(\frac{P_{\text{net}}}{1000}\right)^{0.5} \tag{59}$$

where $C_{\text{owner}}(f)$ is a fuel-dependent base cost (M$ at 1 GWe reference, 2023 USD) and the exponent 0.5 captures staffing economy of scale. Table 14 lists the adopted values.

Table 14: Capitalized owner's cost by fuel cycle (1 GWe reference).

Fuel	Staff	Base (M$)	Key drivers
D-T	117	39	Tritium systems training, HP dept, radwaste, remote handling
D-D	94	31	Reduced neutron/tritium scope
D-^{3}He	69	23	Light HP program, minor tritium
p-^{11}B	59	20	Near-industrial, radiation safety officer (RSO) only

5.17.4 Power-Law Scaling

The exponent of 0.5 is derived from S-PRISM/INL staffing data in section 5.20. Both accounts are driven by total staff count and share the same exponent.

5.17.5 Comparison with Prior Methods

pyFECONS uses the same LSA factor (`fac_91`) for both CAS32 (construction supervision) and CAS40 (owner's costs), fundamentally different scopes. This artifact of the old ARIES Account 91 mapping produces CAS40 ≈ 11–15% of TDC, implicitly assuming fission-level staffing (500 per GWe). The staffing-based approach adopted here yields 1–2% of TDC for fusion, correctly reflecting the 4–8× reduction in headcount under Part 30 regulation.

5.18 CAS50: Capitalized Supplementary Costs

CAS50 covers capitalized items beyond direct construction, indirect services, and owner's costs that must be incurred before commercial operation. CAS50 is decomposed into six sub-accounts, three of which are fuel-dependent:

$$\text{CAS50} = \underbrace{C_{51} + C_{53} + C_{54}}_{\text{fuel-independent}} + \underbrace{C_{52} + C_{55} + C_{56}}_{\text{fuel-dependent}} + C_{59} \tag{60}$$

5.18.1 Fuel-Independent Sub-Accounts

- **C51 (Shipping/transportation):** 1.5% of CAS20. World Nuclear Association reports transportation at 2% of overnight cost for fission [48]; fusion avoids the extreme mass of reactor pressure vessels and containment structures.
- **C53 (Taxes):** 1% of CAS20. Sales and use tax on construction materials, net of exemptions typical for large energy projects (e.g. NY RPTL §485, state manufacturing equipment exemptions).
- **C54 (Construction insurance):** 1.5% of (CAS20 + CAS30). Builder's risk premiums for large projects are typically 1–3% of insured value. Fusion under Part 30 avoids Price-Anderson nuclear liability insurance required for Part 50 reactors.

5.18.2 Fuel-Dependent Sub-Accounts

- **C52 (Spare parts):** Initial on-site inventory at COD. D-T plants require spare blanket modules, first-wall tiles, and divertor cassettes due to neutron damage; p-^{11}B plants need only conventional industrial spares. Expressed as a fraction of CAS22–28 (reactor plant equipment plus BOP).
- **C55 (Startup fuel/inventory):** The fusion analogue of the fission "first core" (3% of overnight cost, \$120M for a 1 GWe PWR). D-T requires 1–2 kg of tritium at \$30,000/g (\$40M); p-^{11}B requires only commodity hydrogen and boron (<\$0.1M).
- **C56 (Decommissioning provision):** The present value of the future decommissioning obligation, discounted over the plant lifetime. See below.

C59 (contingency) is applied at the same rate as other accounts.

5.18.3 Decommissioning Cost Analysis

Fission decommissioning costs \$750–1,250/kWe (OECD-NEA median \$1,000/kWe) [49], driven by spent fuel management, long-lived actinide disposal, and Part 50 regulatory overhead. Fusion eliminates all three: no spent fuel exists, activated materials decay to clearance levels within 50–100 years [50], and Part 30 decommissioning requirements are far lighter.

Total decommissioning cost is estimated by starting from the fission reference and removing inapplicable cost drivers (spent fuel \$300/kWe, actinide waste \$200/kWe, Part 50 overhead \$100/kWe, armed security \$100/kWe), then adding fusion-specific costs (tritium decontamination for D-T). The capitalized provision is the present value at a 3% real discount rate over a 40-year plant life (PV factor ≈ 0.31).

Table 15 summarizes the adopted values.

Table 15: CAS50 fuel-dependent parameters (1 GWe reference, 2023 USD).

Sub-account	D-T	D-D	D-^{3}He	p-^{11}B
C52 spare parts (% of CAS22–28)	3%	2.5%	1.5%	1%
C55 startup fuel (M\$)	40	0.1	10	0.1
C56 decom. provision (M\$)	127	93	65	53
CAS50 total at 1 GWe (M\$)	*493*	*391*	*336*	*293*
% of CAS20	*12.3*	*9.8*	*8.4*	*7.3*

5.18.4 Comparison with Prior Methods

The D-T CAS50 estimate of 8% of total capital is consistent with the Woodruff MIF example at 10% [4]; the p-^{11}B estimate of 5% reflects reduced spare parts, negligible fuel cost, and lighter decommissioning.

5.19 CAS60: Interest During Construction

Interest during construction (IDC) represents the cost of financing the plant during the construction period, as introduced in section 3.2.

5.19.1 Formula

The uniform-spending IDC formula is derived in section 3.2 (eq. (5)); CAS60 reuses it directly:

$$\mathrm{IDC} = C\left[\frac{(1+i)^{T_c}-1}{i\,T_c} - 1\right] = f_{\mathrm{IDC}}(i, T_c) \times C \tag{61}$$

At reference conditions ($i = 0.07$, $T_c = 6$ yr), $f_{\mathrm{IDC}} = 0.192$.

5.19.2 Interaction with CAS90

CAS60 and CAS90 must use complementary conventions to avoid double-counting construction-period financing. Two valid approaches exist:

- **Option A** (adopted): Explicit IDC in CAS60, plain CRF in CAS90.
$$\text{Total capital} = C + \mathrm{IDC}, \quad \mathrm{CAS90} = \mathrm{CRF} \times \text{Total capital} \tag{62}$$
- **Option B**: No CAS60, effective CRF in CAS90.
$$\mathrm{CAS90} = \mathrm{CRF} \times (1+i)^{T} \times C \tag{63}$$

Option A is adopted because it makes IDC visible as a separate line item, uses the more realistic uniform-spending assumption (versus lump-sum in Option B), and is consistent with EMWG/ARIES cost reporting conventions [3].

5.20 CAS70: Annualized O&M and Replacement

CAS70 comprises two sub-accounts:

- **CAS71**: Annual operations and maintenance, scaled from a fuel-dependent base cost per MW of net electric capacity.
- **CAS72**: Annualized scheduled component replacement. Replacement costs are discounted to present value at the time of each replacement event and annualized via the CRF. For the steady-state magnetic-confinement concepts (tokamak, stellarator, mirror, steady-state FRC) the core first-wall and blanket are replaced on a neutron-fluence schedule: the full-power-year lifetime is the per-fuel fluence limit divided by the neutron wall loading,
$$L_{\mathrm{core}} = \mathrm{clamp}\left(\frac{\Phi_{\max}(f)}{q_{\mathrm{wall}}},\ L_{\mathrm{floor}},\ L_{\mathrm{plant}}\right), \tag{64}$$
with $\Phi_{\max}$ the fuel-dependent fluence limit (D-T 18, D-D 36, D-^{3}He 108, p-^{11}B 180 MW yr/m^2; D-T anchored to the ARIES FS 200 dpa RAFM-steel window, the others scaled by spectrum hardness) and q_{wall} the neutron wall loading of eq. (95). A higher-q_{wall} machine wears its core out sooner and carries a larger annualized replacement charge; the lifetime is clamped between a 0.5 FPY floor (keeping the $1/q_{\mathrm{wall}}$ gradient finite at extreme wall loading) and the plant full-power life (nothing is replaced beyond it). Limits and sources are documented in `docs/account_justification/wall_limits_and_fluence.md`. For pulsed inductive DEC concepts, CAS72 includes capacitor bank scheduled replacement, whose lifetime is derived from cycle count and repetition rate. For the electromagnetic-gun concepts (sheared-flow Z-pinch and plasma jet), CAS72 also includes formation-electrode replacement: the plasma-facing coaxial-gun electrodes erode under high current density. Because their shot lifetime can imply sub-annual replacement, this term is taken as a levelized annual recurring cost, $c_{\mathrm{rep}}\,C_{220104}\,N_{\mathrm{shots/yr}}/N_{\mathrm{life}}$, rather than as discrete events. Here $c_{\mathrm{rep}} = 0.5$ is the consumable-electrode share of the driver capital C_{220104} (range 0.25–0.75), $N_{\mathrm{life}} = 10^8$ shots is the electrode lifetime before replacement (range 10^7–10^9, high uncertainty with no NOAK data), and $N_{\mathrm{shots/yr}} = f_{\mathrm{rep}} \times 8760 \times 3600 \times a$ is the annual shot count at repetition rate f_{rep} and availability a; the term scales with the module count n_{mod}.

5.20.1 CAS71: Staffing-Based O&M

The CAS71 O&M coefficient is derived from a bottom-up staffing model that estimates headcount and compensation by division (operations, maintenance, administration, technical, offsite) for each fuel type, then adds non-labor costs (maintenance materials, insurance, supplies, regulatory fees, waste treatment, and general overhead). The staffing model is grounded in two reference datasets: the S-PRISM sodium fast reactor staffing estimate [51] (494 staff for a 1,520 MWe fission plant) and INL's first-principles SFR cost estimation [47], which scales the S-PRISM baseline to different plant sizes. Conventional gas CCGT staffing data (25–35 staff for 1 GWe) anchors the lower bound.

The key differentiator between fuels is neutron-related operational overhead. All fusion machines are regulated under 10 CFR Part 30 (byproduct material), not Part 50 (reactor licensing), eliminating the armed security force, licensed operator requirements, emergency planning zones, and nuclear QA programs that drive fission plant staffing. However, fuels with higher neutron production require radiation protection programs, radwaste management, and radiation-controlled maintenance, costs that scale roughly with neutron flux. The resulting fuel-dependent coefficients are listed in table 16.

Table 16: CAS71 staffing-based O&M coefficients by fuel cycle (1 GWe reference, 2023 USD).

Fuel	Staff	O&M (M$/yr)	O&M (k$/MW/yr)	Key drivers
D-T	117	52.0	52	Tritium systems, HP dept, radwaste, remote handling
D-D	94	38.7	39	Reduced neutron flux ($\frac{1}{3}$ D-T), smaller T inventory
D-^{3}He	69	26.0	26	5% neutron fraction, minimal tritium
p-^{11}B	59	23.6	24	Aneutronic, no tritium, RSO only

All values are at a 1 GWe reference in 2023 USD. For comparison, S-PRISM fission O&M is $50k/MW/yr and modern gas CCGT is $12k/MW/yr. The pB11 estimate ($24k/MW/yr) sits at roughly twice the conventional CCGT level, reflecting fusion-specific system complexity (vacuum, magnets, plasma-facing components) rather than regulatory burden.

Annual O&M scales with plant size via a power law and is modulated by a concept-dependent factor reflecting maintenance ergonomics:

$$\text{CAS71}_{\text{annual}} = C_{\text{O\&M}}(f) \cdot S_{\text{O\&M}}(c) \cdot \left(\frac{P_{\text{net}}}{1000}\right)^{0.5} \tag{65}$$

where the exponent 0.5 captures staffing economy of scale. The INL SFR data [47] shows staff scaling from 236 at 165 MWe to 1,040 at 3,108 MWe ($\alpha \approx 0.5$), driven by the large fixed component in administration, technical, and offsite divisions. The same exponent is used for CAS40 (section 5.17), which shares the same staffing basis.

The concept factor $S_{\text{O\&M}}(c)$ is 1.0 for toroidal geometries (tokamak, stellarator) and 0.85 for linear/open-end geometries (mirror). The fuel-baseline numbers in table 16 are calibrated to a toroidal device; mirrors require fewer maintenance and operations FTEs because blanket rings and first-wall components can be exchanged axially without re-establishing toroidal vacuum and structural continuity, scheduled outages are shorter, and the planned-maintenance crew rotation is smaller. The 0.85 scale is conservative because health physics, tritium accountability, engineering, and security staffing are fuel-driven and concept- agnostic; the accessible concept-specific savings are concentrated in maintenance and hot-cell-operator categories. The same maintenance ergonomics drive the 0.55× capex scaling on CAS22.01.10 remote handling (section 5.9) and the 0.87 default availability for mirrors versus 0.85 for toroidal concepts.

5.20.2 Levelization

Both sub-accounts are levelized using the procedure of section 3.3; neglecting the growing annuity would underestimate by the percentage given there.

5.21 CAS80: Annualized Fuel Cost

CAS80 covers the annualized cost of fuel isotope consumables. The annual fuel expenditure is computed from the fusion power, the energy per reaction, and the unit cost of fuel isotopes:

$$A_{\text{fuel}} = N_{\text{mod}} \cdot P_{\text{fus}}\,[\text{W}] \cdot \frac{8760 \times 3600 \cdot f_{\text{avail}} \cdot c_{\text{rxn}}}{Q_{\text{eff}} \cdot e_{\text{MeV}}} \tag{66}$$

where P_{fus} is in watts, c_{rxn} is the cost per fusion reaction (summing the unit costs of the consumed isotopes weighted by their per-reaction masses), Q_{eff} is the effective energy per reaction in MeV (fuel-dependent; see table 3), and $e_{\text{MeV}} = 1.602 \times 10^{-13}$ J/MeV. The factor 8760×3600 converts one year to seconds. Fuel-specific unit costs are listed in table 11.

5.21.1 Target Consumable for Inertial and Magneto-Inertial Concepts

For inertial and magneto-inertial concepts the fuel-bearing consumable is not only the isotope but the fabricated target destroyed on every shot: the capsule and hohlraum of a laser or heavy-ion target, the beryllium liner and recyclable transmission line of a MagLIF or Z-pinch load. Following the fission convention—where CAS80 holds the fabricated fuel *assembly* (cladding and structure included), not only the raw isotope—this hardware is added to CAS80 as a per-shot term parallel to eq. (66):

$$A_{\text{target}} = N_{\text{mod}} \cdot \dot{N}_{\text{shot}} \cdot c_{\text{target}}, \qquad \dot{N}_{\text{shot}} = f_{\text{rep}} \cdot 8760 \times 3600 \cdot f_{\text{avail}}, \tag{67}$$

where c_{target} is the per-shot target cost (one target per shot per module) and f_{rep} is the repetition rate. The burn-fraction correction below does not apply to A_{target}: the target is consumed whole each shot regardless of fuel burnup. c_{target} is a per-concept input, zero for magnetic concepts and for in-situ-formation pulsed concepts (section 5.3), and it carries only the *non-capital* per-shot cost—materials, factory operating labour, and post-shot radiological handling, divided by the acceptance yield—because the factory capital is funded once, in C_{220108}. This separation matters: published NOAK target costs ($0.20–0.30/shot [28]) are fully-loaded values that already amortize the plant, so charging them in CAS80 *and* a standalone factory capital double-counts the factory. Disentangled and built up from first principles—sub-cent materials (a diamond ablator shell, or a $\sim$3 g lead hohlraum) marked up by tritium-confined handling and divided by yield—the per-shot cost is $\sim$\$0.50–0.62 for laser and heavy-ion capsules and several dollars for the liner-plus-transmission-line loads (the transmission line being net of in-situ recycling). A subtlety the prior treatment omitted is that all post-shot debris is tritiated and neutron-activated, so it must be either recovered in a contained hot cell or conditioned and disposed as low-level waste; for the LIFE-standard lead hohlraum the recovered metal is worth less than the recovery facility, so single-use disposal is the default and is competitive with recycling. Because the rep rate enters linearly, a high-rep laser plant at $f_{\text{rep}} = 10$ Hz consumes $\sim 2.7 \times 10^8$ targets per year, so even a sub-dollar capsule contributes a multiple-percent share of LCOE; once the factory capital and these radiological premiums are costed honestly rather than amortized into an optimistic sub-dollar number, the high-rep capsule concepts sit above their magnetic counterparts in LCOE, as the inertial-fusion economics literature expects.

5.21.2 Burn Fraction and Fuel Recovery

Equation (66) implicitly assumes that all injected fuel undergoes fusion. In practice, only a fraction β_{burn} of the injected fuel fuses per plasma pulse or confinement cycle; the remainder must be exhausted, processed, and either recycled or repurchased. The fuel recovery fraction η_{rec} captures the efficiency of recycling unburned fuel from the exhaust stream. The effective annual fuel cost becomes:

$$A_{\text{fuel}}^{\text{eff}} = A_{\text{fuel}} \left[1 + \frac{1 - \beta_{\text{burn}}}{\beta_{\text{burn}}} (1 - \eta_{\text{rec}}) \right] \tag{68}$$

The multiplier equals unity when either $\beta_{\text{burn}} = 1$ (complete burn) or $\eta_{\text{rec}} = 1$ (perfect fuel recovery). For inexpensive fuels (deuterium at $2,175/kg), the correction is negligible. For expensive fuels (particularly He-3 at $2,000,000/kg), the multiplier is economically significant: at $\beta_{\text{burn}} = 0.10$ and $\eta_{\text{rec}} = 0.95$, the effective fuel cost is $1.45\times$ the full-burn estimate; at $\beta_{\text{burn}} = 0.05$ and $\eta_{\text{rec}} = 0.90$, it rises to $2.9\times$.

Values are declared per concept. The MFE regime default is $\beta_{\text{burn}} = 0.05$, reflecting typical MCF plasma conditions where single-pass burn fractions of 1–10% are expected; inertial and magneto-inertial concepts use individual values in the 0.01–0.30 range. $\eta_{\text{rec}} = 0.99$ is uniform across concepts, reflecting modern exhaust fuel processing. At the MFE-class values the multiplier is $1 + 19 \times 0.01 = 1.19$.

5.21.3 Levelization

CAS80 is levelized using the same growing-annuity procedure as CAS70 (section 3.3).

5.22 CAS90: Annualized Financial Costs

CAS90 converts the total capital investment into an equivalent annual payment:

$$\text{CAS90} = \text{CRF}(i, n) \times \text{Total capital} \tag{69}$$

where total capital includes IDC (CAS60). This uses the plain CRF (eq. (3)), not the "effective CRF" CRF $\times (1+i)^{T_c}$ found in some references [2]. As discussed in section 5.19, the effective CRF is a valid convention only when CAS60 is zero; combining it with an explicit CAS60 double-counts construction financing.

In a constant-dollar framework, CAS91 (operating-period escalation) is zero by convention, and CAS92 (annual regulatory fees) is not modeled. The framework thus equates CAS90 with CAS93 (cost of money).

6 Benchmarking and Cross-Validation

`1costingfe` is exercised against two published reference designs, ARC [31] and ARIES-AT [52], as a calibration cross-check. The framework's contribution is procurement-grounded re-derivation of the cost accounts most sensitive to fuel cycle and confinement family; divergence from heritage-scaling estimates is therefore expected, and the check is whether the divergences fall in the accounts where the methodological change predicts they should. All published costs are escalated to 2025 USD via the BLS Consumer Price Index for All Urban Consumers (CPI-U), matching the convention used by pyFECONS [4]. The Najmabadi 2006 cost basis is in 1992 USD (paper p. 17, "ARIES-AT power-plant economic parameters (1992 \$)"); the Sorbom 2015 cost basis is in FY2014 USD (paper p. 25, Section 6.6). The CPI-U escalation factors used are 2.295 for ARIES-AT (1992 USD $\rightarrow$ 2025 USD) and 1.360 for ARC (2014 USD $\rightarrow$ 2025 USD).

6.1 ARC: Top-Level Comparison

ARC [31] is a 270 MWe demountable-coil high-field tokamak with REBCO HTS magnets and a FLiBe blanket. The inputs used are summarized in table 17.

Table 17: ARC inputs to `1costingfe`.

Parameter	Value	Notes
R_0	3.3 m	major radius
a	1.13 m	minor radius
κ	1.84	elongation
B_0	9.2 T	on-axis field
P_{fus}	525 MW	fusion power
P_{net}	270 MWe	net electric
η_{th}	0.40	He-cooled
Blanket	FLiBe (molten salt)	immersion; drives volume-based CAS27
Fuel	D-T	
Coils	REBCO HTS, demountable	

The published fabricated-component cost is \$5,560M (FY2014 USD, [31] p. 25, Section 6.6), or \$7,562M after escalation to 2025 USD. `1costingfe` predicts an overnight capital cost of \$2,808M, an overnight specific cost of 10,400 \$/kWe, and an LCOE of 145.4 \$/MWh under the framework's NOAK default assumptions. The implied delta on overnight cost is $-63\,\%$. Sorbom's volumetric estimate uses a single \$1.06M/tonne fabricated multiplier averaged across four heritage burning-plasma designs (FIRE, BPX, PCAST5, ARIES-RS), with the dominant contribution from a 4,350-tonne SS316LN structural support priced at \$4.6 B. `1costingfe` replaces this with bottom-up CAS21 (buildings, fuel-cycle-dependent scoping) and CAS22 (procurement-grounded magnet, vessel, and shielding costs); the gap is consistent with replacing a heritage burning-plasma multiplier with current procurement data and reflects a NOAK assumption not present in Sorbom's estimate.

6.2 ARIES-AT: Per-Account Cross-Walk

ARIES-AT [52] is a 1 GWe advanced-tokamak plant with low-temperature superconducting Nb_3Sn coils, a SiC-composite He-cooled Brayton power cycle, and a published per-account cost table that this section uses as the published-side reference. The inputs used are summarized in table 18.

Table 19 compares published top-level costs (escalated to 2025 USD) with `1costingfe`'s prediction. The Najmabadi 2006 paper publishes only the EMWG-style top-level rollups (account 90 "Di-

Table 18: ARIES-AT inputs to `1costingfe`.

Parameter	Value	Notes
R_0	5.2 m	aspect ratio 4.0
a	1.3 m	
κ	2.2	
B_0	5.86 T	on-axis
P_{fus}	1755 MW	
P_{net}	1000 MWe	
η_{th}	0.59	SiC composite, Brayton
Fuel	D-T	
Coils	LTS Nb_3Sn	

rect cost", account 94 "Owner's cost", account 99 "Total cost"), not a full per-CAS-account decomposition; the cross-walk is therefore at the top-level granularity. Direct cost is mapped to the sum of CAS21+CAS22+CAS23+CAS24+CAS25+CAS26+CAS27. The 1costingfe "Overnight" row reports CAS10+CAS21+CAS22+CAS23+CAS24+CAS25+CAS26+CAS27+CAS28+CAS29+CAS30+CAS40+CAS50 (everything before interest during construction); the 1costingfe "Total" row also includes CAS60 (IDC).

Table 19: ARIES-AT cross-walk against published top-level rollups. Published costs from Najmabadi and The ARIES Team [52] Table on p. 17, escalated from 1992 USD to 2025 USD by a factor of 2.295.

Account	Published (2025 USD)	`1costingfe`	Δ (%)
Direct cost (Account 90), M$	3,491	3,488	-0.1
Overnight specific cost, $/kWe	5,602	4,580	-18.2
Total cost (Account 99), M$	6,527	5,461	-16.3
LCOE, $/MWh	109.1	77.2	-29.2

The direct-cost row matches almost exactly (within 0.1 %), indicating that the bottom-up CAS21–27 procurement basis lands on the heritage 2002 ARIES systems-study estimate at the top level. The overnight specific cost is 18 % lower; this is consistent with `1costingfe`'s NOAK default assumption replacing the FOAK-with-learning-multiplier convention used in the ARIES program. The LCOE divergence (-29 %) is dominated by the same NOAK effect propagated through the capital recovery factor.

6.3 LCOE Composition

Figure 4 shows capital cost in M$ broken out by major CAS account group for both reference designs side by side, computed under the framework's NOAK default assumptions.

Both ARC and ARIES-AT produce broadly similar capital stacks dominated by CAS22 (reactor plant equipment), followed by the aggregated indirect and owner's costs (the "Other capital" group, which bundles CAS30 indirect services) and the CAS60 financial charge. CAS21 (buildings) is the largest single direct-equipment account after the reactor, though smaller than those aggregated indirect and financial groups in both cases. The most informative compositional difference is the absolute scale: ARC's compact 270 MWe footprint produces a smaller absolute capital stack than the 1 GWe ARIES-AT but a higher $/kWe figure, reflecting diseconomies of scale in CAS21 buildings and CAS30 indirect services that do not shrink linearly with plant size. The $\eta_{\text{th}} = 0.40$ vs. 0.59 thermal-cycle gap shifts gross-electric recirculation but does not materially shift the capital-stack composition shown here, since fig. 4 reports capital cost rather than LCOE per MWh.

7 Conclusion

A parsimonious differentiable framework for estimating the levelized cost of electricity from fusion power plants has been described. The framework spans the four candidate fuel cycles and the principal confinement families through a common power-balance and cost-account backbone, with each account justified independently from procurement and benchmark data rather than from heritage scaling factors alone. The differentiable backend exposes exact gradients and supports vectorised sensitivity sweeps, making LCOE-driven design exploration tractable.

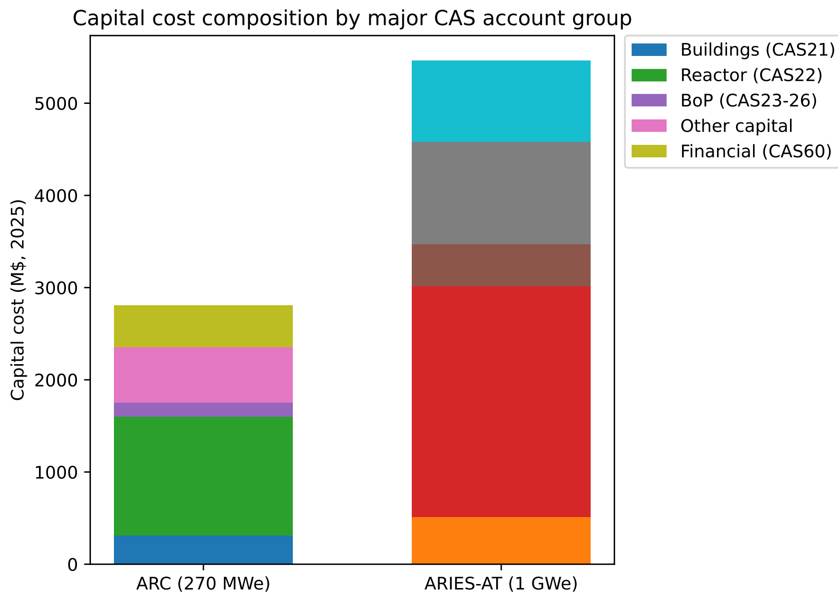

Figure 4: Capital cost composition by major CAS account group for ARC and ARIES-AT, computed by `1costingfe` under NOAK defaults. Annualised costs (CAS70 O&M, CAS80 fuel, CAS90 financial) are not included in this view; see table 19 for the full LCOE breakdown. All values in 2025 USD.

Several limitations remain. Only a single confinement-specific physics layer (the 0D tokamak model of appendix A) is implemented; analogous models for stellarators, mirrors, FRCs, and inertial systems are planned but not yet exercised. Calibration cross-checks against the ARC and ARIES-AT reference designs are presented in section 6; cross-validation against pyFECONS reference cases at the per-account level is left to future work.

Acknowledgments

This work was performed within the 1cFE program at the Astera Institute Residency. The 1cFE program seeks to identify plausible corridors to fusion electricity at $\leq$ \$0.01/kWh (2025 USD); see https://1cf.energy/. `1costingfe` is the cost-accounting layer of the program's broader techno-economic analysis stack, developed alongside the `fusion-tea` repository family.

8 Code Availability

The `1costingfe` source code, example scripts, and per-account documentation are released at https://github.com/1cfe/1costingfe. The 1cFE program landing page is available at https://1cf.energy/. The framework targets Python 3.11 or later and depends on JAX. Reproduction of most figures and account-level cost tables in this paper requires only the example scripts shipped in the `examples/` directory; the benchmarking results in section 6 are recorded by the scripts `benchmark_arc.py`, `benchmark_aries_at.py`, and `make_benchmark_bars.py`, located in `docs/papers/1costingfe_paper/scripts/`. The section 2 walkthrough is recorded by `examples/external_physics_handoff.py` (the rollup table) and `docs/papers/1costingfe_paper/scripts/make_tornado.py` (the sensitivity tornado figure).

A 0D Tokamak Model

The power balance of the physics module accepts fusion power P_{fus} as an input. To derive P_{fus} from machine geometry and plasma parameters for a tokamak, the framework implements a zero-dimensional (0D) equilibrium model that chains standard scaling laws. The model is fully differentiable, enabling gradient-based sensitivity analysis of LCOE with respect to machine-level design choices (R, a, B, q_{95}, etc.).

A.1 Fusion Reactivity

The fusion reactivity $\langle\sigma v\rangle$ is evaluated per fuel with analytic parameterizations of the common form

$$\langle\sigma v\rangle = C_1\,\theta\,\sqrt{\frac{\xi}{m_r c^2\,T^3}}\;\exp(-3\xi) \quad [\mathrm{m^3/s}] \tag{70}$$

where $\theta = T/(1 - T(C_2 + T(C_4 + TC_6))/(1 + T(C_3 + T(C_5 + TC_7))))$, $\xi = (B_G^2/4\theta)^{1/3}$, B_G is the Gamow constant of the reaction, and C_1–C_7 are tabulated fit coefficients. D-T ($B_G = 34.38$ keV$^{1/2}$, valid $0.2 \le T \le 100$ keV, peak near $T \approx 65$ keV), both D-D branches, and D-^{3}He (valid to 190 keV) use the coefficients of Bosch and Hale [9]; p-B11 uses the high-temperature fit of Nevins and Swain [53] (valid $50 \le T \le 500$ keV) as tabulated by Tentori and Belloni [54], with the higher-reactivity revision of Putvinski et al. [12] available as the optimistic alternative.

Reactant densities follow quasineutrality from the electron density and a fuel-mix ratio input ($n_{^3\mathrm{He}}/n_\mathrm{D}$ or $n_\mathrm{B}/n_\mathrm{p}$): the higher fuel-ion charges of the advanced fuels dilute the reactant densities, reduce the total ion pressure entering the stored energy and β_N, and raise Z_eff (and with it bremsstrahlung), with the configured Z_eff interpreted as the hydrogenic-plus-impurity baseline. For D-^{3}He the D-D side-channel fraction is derived from the rate ratio at the operating point (an explicit input pins it); the ion temperature can exceed the electron temperature through a hot-ion ratio $T_i/T_e \ge 1$; and the operating-temperature search interval is fuel-specific, bounded by each fit's validity range. For D-^{3}He and p-B11 the radiated power defaults to the same fixed-fraction-of-fusion proxy used by the energy-balance path, keeping the two paths consistent.

A.2 Plasma Equilibrium

Given major radius R, minor radius a, elongation κ, toroidal field B, and edge safety factor q_{95}, the plasma current follows from MHD force balance:

$$I_p = \frac{2\pi\,a^2\,\kappa\,B}{\mu_0\,R\,q_{95}} \quad [\mathrm{A}] \tag{71}$$

The electron density is set as a fraction f_GW of the Greenwald density limit [55]:

$$n_\mathrm{GW} = \frac{I_p\,[\mathrm{MA}]}{\pi\,a^2} \quad [10^{20}\ \mathrm{m^{-3}}], \qquad n_e = f_\mathrm{GW}\,n_\mathrm{GW} \tag{72}$$

The plasma volume and first-wall area of the elongated torus are $V = 2\pi^2 R\,a^2\,\kappa$ and $A_\mathrm{fw} = 4\pi^2 R\,a\,\kappa$.

A.3 Fusion Power

For a 50/50 D-T plasma ($n_D = n_T = n_e/2$), the volumetric fusion power is:

$$P_\mathrm{fus} = \tfrac{1}{4}\,n_e^2\,\langle\sigma v\rangle(T_e)\;E_\mathrm{fus}\,V \quad [\mathrm{W}] \tag{73}$$

where $E_\mathrm{fus} = 17.6$ MeV per D-T reaction. This P_fus feeds directly into the fuel-cycle-dependent power split (table 3) and the power balance of eq. (26)–eq. (30).

A.4 Energy Confinement

The energy confinement time is evaluated against the IPB98(y,2) H-mode scaling [55]:

$$\tau_{E,98} = 0.0562\;I_p^{0.93}\,B^{0.15}\,\bar{n}_{19}^{0.41}\,P_\mathrm{heat}^{-0.69}\,R^{1.97}\,\epsilon^{0.58}\,\kappa^{0.78}\,M^{0.19} \quad [\mathrm{s}] \tag{74}$$

where $\bar{n}_{19}$ is the line-averaged density in units of 10^{19} m^{-3}, $P_\mathrm{heat} = P_\alpha + P_\mathrm{aux}$ is the total heating power in MW, $\epsilon = a/R$ is the inverse aspect ratio, and M is the effective ion mass in AMU. The ratio of the actual confinement time $\tau_E = W_\mathrm{th}/P_\mathrm{heat}$ (where $W_\mathrm{th} = 3\,n_e\,T_e\,V$ for electrons and ions at equal temperatures) to the scaling prediction gives the H-factor, $H = \tau_E/\tau_{E,98}$, which measures the quality of confinement relative to the empirical database.

A.5 Stability and Engineering Limits

The normalized beta,

$$\beta_N = \frac{\beta_t \, a \, B}{I_p \, [\text{MA}]}, \qquad \beta_t = \frac{2\mu_0 \, n_e \, T_e}{B^2}, \tag{75}$$

must remain below the Troyon limit ($\beta_N \lesssim 3.5$) for ideal MHD stability. The safety factor q_{95} must exceed 2 to avoid kink instabilities.

Two engineering constraints are evaluated: the neutron wall loading $q_{\text{wall}} = P_n / A_{\text{fw}}$, which drives structural damage and blanket lifetime; and the divertor heat flux, estimated from a simplified scrape-off-layer model with midplane power width $\lambda_q \approx 2$ mm and flux expansion factor $f_{\text{exp}} \approx 5$.

A.6 Disruption Model

Disruptions destroy confinement and can cause significant structural damage. The model estimates a disruption frequency from the proximity of the operating point to stability boundaries:

$$\Gamma = \sum_j \Gamma_0 \, \exp(-s \, m_j) \tag{76}$$

where the sum runs over the Greenwald ($m = 1 - f_{\text{GW}}$), Troyon ($m = 1 - \beta_N / \beta_{N,\max}$), and kink ($m = 1 - q_{95,\min}/q_{95}$) channels, $\Gamma_0 = 0.1$ FPY^{-1} is the baseline rate far from limits, and $s = 15$ sets the steepness. Each disruption damages a fraction of the first-wall component lifetime and incurs 72 hours of downtime, reducing effective plant availability.

A.7 Operating Modes

The model supports two operating modes. In *forward mode*, the user specifies machine geometry, field, safety factor, Greenwald fraction, and temperature; the model returns the full plasma state including P_{fus}, β_N, H-factor, wall loading, and disruption rate. In *inverse mode*, the user specifies a target P_{net} and the model solves for the electron temperature T_e that produces the required fusion power via bisection, returning both the plasma state and the complete power table. The inverse mode enables target-driven design: given a net power requirement, the framework determines the plasma conditions, checks them against stability and engineering limits, and computes the resulting LCOE in a single differentiable pass. The implied operating point must be admissible: a violation of an error-severity stability limit (Greenwald fraction, Troyon β_N, kink safety factor) raises an `OperatingPointInfeasible` error carrying the diagnosis, rather than costing a plasma that cannot exist; an explicit flag (`enforce_plasma_limits`) disables the check for exploratory runs. Far beyond the limits the disruption-rate model is numerically capped and the disruption-penalized availability floored at zero, so no operating point can produce non-finite cost components.

A.8 Power-to-Geometry Sizing

For cost analyses that must scale with the power target, the framework provides a sizing solve that maps a target net electric power to tokamak geometry (R_0, a, B_0), driving the geometry-dependent accounts (C220103 coils, blanket, vessel) from physics rather than from pinned inputs.

The on-axis toroidal field is derived from the conductor's peak-field ceiling $B_{\max}$ and the inboard radial build:

$$B_0 = B_{\max} \frac{R_{\text{coil,i}}}{R_0}, \qquad R_{\text{coil,i}} = R_0 - a - d_{\text{in}}, \tag{77}$$

where $d_{\text{in}} = d_{\text{bk}} + d_{\text{sh}} + d_{\text{str}} + d_{\text{vv}}$ is the sum of the inboard blanket, shield, structure, and vessel thicknesses. Because these layers are fixed in metres, the ratio $B_0/B_{\max}$ falls as R_0 decreases, making high-field conductor more advantageous at small machine size.

At each trial R_0 the operating point is determined at the constraint boundary: plasma current from eq. (71) with the q_{95} floor, density from eq. (72) at the Greenwald fraction f_{GW}, and temperature from a bounded maximization of net power over a per-fuel interval $[T_{\min}, T_{\max}]$, subject to the Troyon limit $\beta_N \leq \beta_{N,\max}$. Net power is then evaluated with the full power balance, including resistive recirculation for copper coils. Net power is monotonic increasing in R_0 at the boundary operating point, so R_0 is located by bisection; if the maximum search radius $R_{0,\max}$ cannot deliver the target, the solve raises a `SizingInfeasible` error.

The solve supports four usage modes: *pin* (existing behavior, sizing off), *size* (solve R_0 from design knobs), *scale* (anchor a reference machine's knobs, change the power target), and *optimize* (minimize LCOE over f_{GW} with R_0 slaved to the power target, using the disruption model of appendix A). Both sizing flags default to false, so existing configurations are unaffected. The solve is fuel-aware through the per-fuel reactivities, mix dilution, and radiation treatment of appendix A.1, with fuel-specific temperature search intervals; sized D-^{3}He machines are larger than D-T at equal net power, and p-B11 does not reach positive net power within the search radius and fails as infeasible.

A.9 Literature Anchors

The tokamak closed energy balance is the template the mirror sustainment model of appendix B.6 copies, so the reference is itself pinned to published design points, parallel to the mirror GDT/WHAM anchors of appendix B.10. Run at each machine's published volume-averaged density and temperature with the published auxiliary power, the unchanged forward model reproduces the published fusion gain Q within a factor 1.09 for ITER [56] and 0.83 for ARC [31], both inside the 2x gate, and within a factor of about three for SPARC [57], the larger gap reflecting SPARC's strongly peaked profiles against the flat-profile 0D idealization. As with the mirror anchors, the 2x band exists for the flat-versus-peaked profile gap rather than a calibration freedom; no tokamak physics or cost coefficient is adjusted to the anchors.

B 0D Mirror Model

The framework includes a zero-dimensional (0D) plasma model for axisymmetric magnetic mirrors, structurally parallel to the tokamak model of appendix A. Given machine geometry (length L, plasma radius a, minimum on-axis field $B_{\min}$, mirror ratio R_m) and plasma conditions (T_i, T_e, n_e), the model computes confinement times, fusion power, plasma beta, and wall loading. The model is fully JAX-differentiable and float32-safe for use in gradient-based analysis.

B.1 Ion-Ion Collision Time

The model uses the NRL ion-ion collision time [8] for a singly charged fuel ion of mass number A and charge $Z = 1$:

$$\tau_{ii} = \frac{2.09 \times 10^{13}\, A^{1/2}\, T_i[\text{eV}]^{3/2}}{n_i\, \ln\Lambda} \quad [\text{s}] \tag{78}$$

where $T_i[\text{eV}]$ is the ion temperature in eV, n_i is the ion density in m^{-3}, and $\ln\Lambda = 17$ is the Coulomb logarithm. The implementation accepts T_i in keV and n_i in units of 10^{20} m^{-3}, folding both conversions into a single prefactor:

$$\tau_{ii} = \frac{(2.09 \times 10^{13})\,(10^3)^{3/2}\,10^{-20}}{\ln\Lambda} \cdot \frac{A^{1/2}\, T_{i,20}^{3/2}}{n_{i,20}} = \frac{6.609 \times 10^{-3}\, A^{1/2}\, T_{i,20}^{3/2}}{n_{i,20}\, \ln\Lambda} \quad [\text{s}] \tag{79}$$

where $T_{i,20} = T_i$ [keV] and $n_{i,20} = n_i/10^{20}$ m^{-3}. This convention keeps all intermediates near unity, avoiding float32 overflow in the XLA compiler.

B.2 Confining Potential: Tandem Plug

The reactor-relevant mirror is a tandem: the central cell that does the fusion is confined by an electrostatic potential drop to high-field end plugs, not by the simple-mirror electron-loss potential. Following Fowler and Logan [58] the central-cell ion confining potential is set by the plug-to-central density ratio n_p/n_c:

$$e\phi = T_{e,\text{plug}}\, \ln\!\left(\frac{n_p}{n_c}\right), \tag{80}$$

where $T_{e,\text{plug}}$ is the plug hot-electron temperature. The plug is a separate cell whose hot-electron population, sustained by plug ECH, holds the confining potential, so $T_{e,\text{plug}}$ is decoupled from the central-cell electron temperature [59]: a thermal barrier lets the plug run hot (default $T_{e,\text{plug}} = 125$ keV, anchored to the high-field tandem design point of Frank et al. [60]) while the central cell sets its own T_e for radiation, beta, and stored energy. Because n_p/n_c is an order-unity-to-few ratio, $e\phi$ is bounded, and at the anchor ($n_c/n_p = 0.55$, $T_{e,\text{plug}} = 125$ keV) it evaluates to $e\phi = 74.7$ keV, matching the published tandem central-cell value.

The simple-mirror Boltzmann ambipolar potential

$$e\phi_{\text{amb}} = T_e \ln\sqrt{\frac{A\, m_p}{2\pi\, m_e}} \tag{81}$$

(about $3.2\, T_e$ for deuterium) is retained as a diagnostic for the simple-mirror limit; the tandem confining potential of eq. (80) is the one entering the confinement chain.

B.3 Confinement Time Channels

The model implements three confinement channels, each physically distinct.

B.3.1 Classical Mirror Confinement (Bing–Roberts)

Ions trapped in a loss-cone distribution scatter into the loss cone on the collision time. Following Huba [8]:

$$\tau_{\text{cl}} = 2.6\,\ln(R_m)\,\tau_{ii} \tag{82}$$

This channel is diagnostic only; it is not combined into τ_p because the Pastukhov and gas-dynamic channels dominate in the parameter space of interest.

B.3.2 Pastukhov Electrostatic Plugging

The ambipolar potential $e\phi$ raises the effective loss-cone barrier. Ions must diffuse in energy past $e\phi + T_i$ to escape, giving an exponential suppression [61, 62]:

$$\tau_{\text{Pas}} = \frac{\sqrt{\pi}}{2}\,\frac{R_m+1}{R_m}\,\ln(2R_m+2)\,\frac{e\phi}{T_i}\,e^{e\phi/T_i}\,\tau_{ii} \tag{83}$$

The factor $\ln(2R_m + 2)$ captures the logarithmic density-of-states weighting over the mirror throat, and $(R_m + 1)/R_m$ is a geometric correction from the full Pastukhov 1974 derivation.

B.3.3 Gas-Dynamic Confinement

When the mean free path is shorter than the device length ($\ell_{\text{mfp}} \lesssim L$), collisions establish a nearly Maxwellian distribution that leaks through the loss cone as a thermal gas. The gas-dynamic confinement time [63] is:

$$\tau_{\text{GD}} = \frac{R_m\,L}{v_{ti}}, \qquad v_{ti} = \sqrt{\frac{2\,T_i\,[\text{keV}]\,e_{\text{keV}}}{A\,m_p}} \tag{84}$$

where $e_{\text{keV}} = 1.602 \times 10^{-16}$ J/keV is the keV-to-J conversion. In the implementation v_{ti} is the ion thermal speed based on T_i; this is a factor of $\sqrt{2}$ larger than the sound speed $c_s = \sqrt{T_e/m_i}$ used in some formulations, an explicit convention documented in the validation anchors (appendix B.10).

B.3.4 Radial Cross-Field Diffusion

Classical diffusion across field lines contributes a radial channel:

$$\tau_{\text{rad}} = \left(\frac{a}{\rho_i}\right)^2 \tau_{ii}, \qquad \rho_i = \frac{\sqrt{2\,A\,m_p\,T_i\,[\text{keV}]\,e_{\text{keV}}}}{e\,B_{\min}} \tag{85}$$

where ρ_i is the ion gyroradius at the midplane field $B_{\min}$.

B.3.5 Combined Axial Confinement: Collisionality Bridge

The Pastukhov and gas-dynamic channels apply in physically distinct collisionality regimes and must be joined by a bridge that selects the governing limit. Rognlien and Cutler [64] establish that the Pastukhov loss-cone formula holds in the collisionless limit, while the collisional gas-dynamic outflow governs once the ion mean free path, reduced by the mirror ratio, falls to the order of the device length; the regime boundary is therefore at collisionality

$$\xi \equiv \frac{L}{v_{ti}\,\tau_{ii}} \sim \frac{1}{R_m}, \tag{86}$$

with $\xi \gg 1/R_m$ collisional and $\xi \ll 1/R_m$ collisionless. The gas-dynamic loss rate physically exists only when the loss cone is kept full by collisions, so it is gated by a smooth logistic in $\log_{10}\xi$ centered on the boundary and combined via the loss-rate sum:

$$g(\xi) = \sigma\!\left(\frac{\log_{10}\xi - \log_{10}(1/R_m)}{w}\right), \tag{87}$$

$$\frac{1}{\tau_{\text{axial}}} = \frac{1}{\tau_{\text{Pas}}} + g(\xi)\,\frac{1}{\tau_{\text{GD}}}, \tag{88}$$

where σ is the logistic function and $w = 0.13$ decades is a constructed smoothing width (Rognlien and Cutler establish the transition is smooth but publish no closed-form width). As $g \to 1$ in the collisional regime the gas-dynamic time dominates and $\tau_{\text{axial}} \to \tau_{\text{GD}}$; as $g \to 0$ in the collisionless regime the gas-dynamic rate is suppressed below the Pastukhov rate and $\tau_{\text{axial}} \to \tau_{\text{Pas}}$. The logistic argument is clamped to $[-30, 30]$ before exponentiation to keep the reverse-mode gradient finite. The total particle confinement combines axial and radial:

$$\frac{1}{\tau_p} = \frac{1}{\tau_{\text{axial}}} + \frac{1}{\tau_{\text{rad}}} \tag{89}$$

B.4 Energy Confinement Time

Particles lost axially carry the ambipolar potential energy $e\phi$ in addition to their thermal energy. The energy confinement time is therefore shorter than the particle confinement time by the ratio of stored thermal energy to total loss energy per particle:

$$\tau_E = \tau_p \cdot \frac{\frac{3}{2}(T_i + T_e)}{2\, e\phi + T_i + T_e} \tag{90}$$

B.5 Alpha Loss-Cone Heating

A mirror does not confine its fusion alphas the way a closed device does: a fraction of the 3.5 MeV alphas is born in or scatters into the velocity-space loss cone and streams out the ends before depositing its energy. Santarius and Callen [65] give a bounce-averaged Fokker-Planck result that roughly half the alphas are lost by count but under a quarter by energy, so $f_{\alpha,\text{heat}} \approx 0.75$–$0.85$ of the charged-particle power deposits in the central cell (model default $f_{\alpha,\text{heat}} = 0.80$). The model credits only $f_{\alpha,\text{heat}}\, P_\alpha$ as self-heating and routes the remaining $(1 - f_{\alpha,\text{heat}})\, P_\alpha$ to the axial end-loss channel as directed loss-cone exhaust, where it is recovered by the end-plug direct converter alongside the thermal end loss; the alpha energy is conserved, not discarded.

B.6 Energy-Balance Closure and Plug Power

In sizing mode the auxiliary sustainment power is charged from the closed steady-state energy balance rather than asserted. The confinement-derived sustainment is

$$P_{\text{aux}} = \max\!\left(P_{\text{aux}}^{\text{floor}},\ P_{\text{end}} + P_{\text{radial}} + P_{\text{rad}} - f_{\alpha,\text{heat}}\, P_\alpha\right), \tag{91}$$

which is fed as the heating input to the shared MFE power balance of appendix D; the transport channel then conserves the undeposited alpha power, $p_{\text{transport}} = P_{\text{end}} + P_{\text{radial}} + (1 - f_{\alpha,\text{heat}})\, P_\alpha$, so the fusion-to-injected gain $Q_{\text{sci}} = P_{\text{fus}}/P_{\text{aux}}$ and the engineering gain follow self-consistently from the confinement. The hot-electron plug is sustained at a real power cost: a plug power P_{plug} (default 30 MW, anchored to the high-field tandem design point of Frank et al. [60]) is charged into the recirculating budget, since the plug ECH heats the plug electrons rather than the central cell. This plug power is the competing penalty that keeps the optimizer honest: deep plug confinement is not free.

Because the Fowler-Logan potential of eq. (80) is fixed by the plug rather than scaling with T_i, the ratio $e\phi/T_i$ falls as the central cell is heated, so the Pastukhov enhancement weakens with T_i: heating the central cell costs confinement. The length-sizing and LCOE-optimize search therefore settles the D-T central cell at a cool, genuinely driven operating point ($T_i \approx 23$ keV) with auxiliary sustainment well above the control floor, rather than running to ignition.

B.7 Plasma Beta and Operating Limits

Midplane plasma beta is computed from the ion and electron pressures at the minimum field B_{min}:

$$\beta = \frac{2\,\mu_0\, n_e\, (T_e + n_i/n_e \cdot T_i)\, e_{\text{keV}}}{B_{\text{min}}^2} \tag{92}$$

where the n_i/n_e dilution factor accounts for higher-charge advanced fuels. An MHD stability constraint $\beta \le \beta_{\text{max}}$ is enforced by `OperatingPointInfeasible` at error severity.

B.8 Inverse Contract and Length Sizing

The inverse mode fixes T_e and n_e as inputs and bisects over $T_i \in [T_{\text{min}}, T_{\text{max}}]$ (fuel-specific) to find the ion temperature that delivers a target net electric power P_{net}^*. A single pass is used: (i) call the MFE inverse power balance with the input direct-energy-conversion fraction f_{dec} to obtain the required fusion power P_{fus}^*; (ii) bisect on T_i until

$P_{\text{fus}}(T_i, n_e, V) = P^*_{\text{fus}}$; (iii) run the full forward model at the solved T_i to build the complete plasma state. The parameter f_{dec} is an engineering input (default 0.30 for D-T), not derived from the plasma model.

For length sizing ($P_{\text{net}} \to L$), the operating density is the minimum of three caps,

$$n_e = \min(n_\beta,\ n_{\text{wall}},\ n_{\text{surf}})\,, \tag{93}$$

each the density at which a distinct first-wall or stability limit binds. The beta-pressure cap sets the density at a fraction f_β of the MHD stability boundary:

$$n_\beta = f_\beta\,\beta_{\max}\,\frac{B^2_{\min}}{2\,\mu_0\,(T_e + n_i/n_e \cdot T_i)\,e_{\text{keV}}} \tag{94}$$

The neutron wall-load cap limits the 14.1 MeV fluence on the first wall. Writing the neutron wall loading as

$$q_{\text{wall}} = \frac{P_{\text{neutron}}}{A_{\text{fw}}}, \qquad A_{\text{fw}} = 2\pi\,(a + t_{\text{vac}})\,L, \tag{95}$$

and using $P_{\text{neutron}} = f_n\,C_{\text{fus}}(T)\,n_e^2\,V$ with the plasma volume $V = \pi a^2 L$ (the L cancels, so the cap is set by a, n_e, and T alone), the wall-cap density follows in closed form,

$$n_{\text{wall}} = \sqrt{\frac{2\,q^{\max}_{\text{wall}}\,(a + t_{\text{vac}})}{f_n\,C_{\text{fus}}(T)\,a^2}}, \tag{96}$$

where f_n is the neutron energy fraction and $C_{\text{fus}}(T)$ the per-n_e^2 fusion power density coefficient. The surface heat-flux cap limits the photon and cross-field charged-particle load deposited in the first microns of the wall,

$$q_{\text{surface}} = \frac{P_{\text{rad}} + P_{\text{radial}}}{A_{\text{fw}}} \le q^{\max}_{\text{surface}}, \tag{97}$$

with the axial end-loss excluded (it exits through the throats to the expander, not the lateral wall). Because radiation plus radial transport is monotone increasing in density at fixed T, n_{surf} is found by a single uniform bisection on n_e for all fuels. The default caps are $q^{\max}_{\text{wall}} = 5$ MW/m^2 and $q^{\max}_{\text{surface}} = 1$ MW/m^2, sourced in the wall-limits provenance writeup. For D-T the neutron cap typically binds first; for advanced fuels the surface cap is the governing wall constraint. At the resolved density $P_{\text{fus}} \propto n_e^2 V \propto n_e^2 L$ is monotonically increasing in L (for fixed n_e), so L is located by bisection on $[L_{\min}, L_{\max}]$. If $L_{\max}$ is insufficient, or if even the capped density cannot reach P^*_{net}, a `SizingInfeasible` error is raised.

B.9 Two-Class Coil Scaling

Mirror coil costs are divided into two classes: (i) central-cell solenoid coils, with continuous count $n_{\text{central}} = L/d_{\text{coil}}$ growing with chamber length at spacing d_{coil} (default 5 m, no rounding so the expression is JAX-differentiable), at the design field b_{center} and a bore $r_{\text{coil}} = r_{\text{vessel,out}} + 0.10$ m taken from the radial build, and (ii) end-plug coils with fixed count n_{plug} (default 4, two per end), at the throat field $R_m B_{\min}$ and a bore $r_{\text{coil}} = a/\sqrt{R_m} + 0.30$ m set by throat flux conservation. The two bores differ in kind: the central solenoids enclose the full plasma–blanket–shield–vessel stack, while the plug coils sit where the plasma necks down and no blanket extends. The total coil cost scales as a linear combination of the two counts, carrying the mirror markup $M = 1.7266$ calibrated at the YAML default chamber length of 20 m ($n_{\text{central}} = 4$, $n_{\text{plug}} = 4$, baseline 513 M$).

B.10 Validation Anchors

The confinement kernels are validated against two published machines (details in the test suite):

GDT (gas-dynamic anchor.) The Gas Dynamic Trap at Budker Institute [66] operates with a 7 m central cell, $R_m = 35$, and deuterium. In the warm-plasma gas-dynamic regime (no NBI, $T_e \approx 0.25$ keV, $n_e \approx 2 \times 10^{19}$ m^{-3}), Endrizzi et al. [67] (eq. 3.5) give $\tau_{\text{GD}} = 5.2\,R_m\,L_p\,T_e^{-1/2}$ μs, where L_p is the half-length. Evaluated at GDT parameters this gives 1.27 ms. The model kernel using the full length and the ion thermal speed gives 1.58 ms; the factor of $\sqrt{2}$ relative to the half-length/sound-speed convention is expected and documented. Ratio 1.25, within the 2x validation gate.

WHAM (Pastukhov anchor.) The Wisconsin HTS Axisymmetric Mirror design [67] has $R_m \approx 19.8$, $n_e = 3 \times 10^{19}$ m^{-3}, and a midplane mean ion energy of 10 keV at beta $= 0.2$. The published Pastukhov/classical-mirror scaling (Endrizzi eq. 3.4) gives $n_{20}\,\tau_p = 250\,E_{b,100}^{3/2}\,\log_{10} R_m$ ms for a 25 keV NBI beam, which evaluates to 135 ms. The model Pastukhov kernel at the same midplane thermal point ($T_i = 10$ keV, $T_e = 1$ keV) gives 88 ms; the ratio 1.53 reflects the expected gap between a single thermal Maxwellian and the Fokker-Planck fast-ion calculation underlying the published scaling. Within the 2x gate. The collisionality bridge of eq. (88) places GDT on the gas-dynamic branch (collisional, ξ above $1/R_m$) and WHAM on the Pastukhov branch (collisionless, ξ more than three decades below $1/R_m$), each within one percent of the governing kernel.

Hammir (tandem anchor.) The high-field axisymmetric tandem design point of Frank et al. [60] has a 50 m central cell, $n_c/n_p = 0.55$, $T_i = 45$ keV, $T_{e,\text{plug}} = 125$ keV, $P_{\text{fus}} = 157$ MW, and 30 MW of plug NBI/ECH, giving a published gain $Q = P_{\text{fus}}/P_{\text{NBI}} \approx 5.2$. At this central cell the tandem confining potential of eq. (80) gives $e\phi = 74.7$ keV, the central cell is not ignited, and the model gain lands within the 2x band; charging $P_{\text{plug}} = 30$ MW makes the engineering-gain accounting match the published Q, which counts the same plug power.

B.11 Headline Operating Point

Sized to a 400 MWe net target at the default high-field tandem ($f_\beta = 0.85$), the D-T mirror is a driven tandem at $T_i = 23$ keV with $Q_{\text{sci}} = 8.3$, energy confinement $\tau_E = 19$ s, auxiliary sustainment near 245 MW, engineering gain $Q_{\text{eng}} = 1.68$, and LCOE about 396 \$/MWh; the operating point is beta-bound. D-T is therefore net-positive but marginal, paying the full recirculating cost of its hot-electron plug. The advanced fuels (D-D, D-^{3}He, p-^{11}B) are each genuinely driven but net-negative at the modelled fields: the hot-electron plug needed to confine the central cell radiates against their lower or harder reactivity, so none reaches $Q_{\text{eng}} > 1$ even with a cool central cell. Only D-T is economic in the 0D model.

B.12 Assumptions and Limitations

The model makes the following approximations, inherited from the 0D design philosophy:

- Single thermal Maxwellian distribution. No fast-ion population, sloshing-ion anisotropy, or NBI slowing-down physics is included. Real mirror devices (GDT, WHAM) are beam-driven; the model is appropriate for a steady-state reactor operating point where the alpha population thermalizes quickly compared to τ_p.
- The central-cell T_e and the plug $T_{e,\text{plug}}$ are inputs; there is no self-consistent electron power balance within the plug.
- f_{dec} is an engineering input, not derived from f_{axial} of the plasma state. The diagnostic field `f_axial_derived` is available for post-processing.
- The plug is modeled as a confining potential plus a fixed sustainment power, not a self-consistent second fusion plasma; the plug density ratio and plug power are inputs anchored to a single published design point.
- MHD stability is enforced only via the $\beta \le \beta_{\max}$ gate. A collisionality-validity flag and a DCLC loss-cone proxy are reported as informational diagnostics, but drift-resonance and mirror-mode instabilities do not restrict the operating point.
- Geometry is a straight cylinder; end-cell and expander regions are not modeled.

C Synchrotron radiation model details

This appendix documents the implementation details of the synchrotron radiation model summarized in equation (19).

C.1 Profile Assumptions and Central-Value Conversion

The Albajar model is a model of synchrotron radiation out of a tokamak plasma, and it requires central (on-axis) values of electron temperature and density, whereas `1costingfe`'s power balance operates on volume-averaged quantities. The assumed radial profiles follow the convention of Albajar et al. [16]:

$$T_e(\rho) = T_{e0}\,(1-\rho^{\beta_T})^{\alpha_T}, \qquad n_e(\rho) = n_{e0}\,(1-\rho^2)^{\alpha_n} \tag{98}$$

where $\rho = r/a$ is the normalized minor radius. For profiles depending only on ρ, the toroidal volume average reduces exactly to $\langle f \rangle = 2\int_0^1 f(\rho)\,\rho\,d\rho$ (the $(R + a\rho\cos\theta)$ Jacobian factor integrates out). For $\beta_T = 2$ (the default) this gives

$$\langle T_e \rangle = \frac{T_{e0}}{1+\alpha_T}, \qquad \langle n_e \rangle = \frac{n_{e0}}{1+\alpha_n} \tag{99}$$

The code inverts these relations to obtain central values from the volume-averaged inputs: $T_{e0} = \langle T_e \rangle\,(1+\alpha_T)$ and $n_{e0} = \langle n_e \rangle\,(1+\alpha_n)$. The temperature profile shape exponent is fixed at $\beta_T = 2$ (standard parabolic profile). Default peaking exponents are $\alpha_T = 1.0$ and $\alpha_n = 0.5$, giving peaking factors of 2.0 and 1.5, respectively.

C.2 Profile Shape Factor

The profile shape factor K in equation (19) is [16], evaluated at $\beta_T = 2$:

$$K(\alpha_n, \alpha_T) = (\alpha_n + 3.87\,\alpha_T + 1.46)^{-0.79}\,(1.98+\alpha_T)^{1.36}\,2^{2.14}\,(2^{1.53} + 1.87\,\alpha_T - 0.16)^{-1.33} \tag{100}$$

C.3 Aspect Ratio Correction

The geometric correction factor G accounts for the finite aspect ratio of the torus [16]:

$$G(A) = 0.93\left[1 + 0.85\,e^{-0.82\,A}\right] \tag{101}$$

where $A = R/a$ is the aspect ratio. In the large aspect-ratio limit $G \to 0.93$.

C.4 Application to Non-Tokamak Geometries

For stellarators the formula is applied directly using the stellarator's major and minor radii. For magnetic mirrors, which have a cylindrical geometry of length L and plasma radius a, an effective major radius is defined as

$$R_{\text{eff}} = \frac{L}{2\pi} \tag{102}$$

This maps the cylinder to a torus of equivalent volume ($V = 2\pi^2 R a^2 \kappa \to \pi a^2 L$ for $\kappa = 1$). The wall reflectivity R_w is reduced for mirrors (default 0.4 vs. 0.6 for tokamaks) to account for radiation escaping through the open ends.

D Inverse Power Balance

The costing model operates in "inverse" mode: given a target net electric output P^*_{net}, determine the required fusion power P_{fus}.

D.1 Pulsed Inverse

For pulsed concepts, Q_{eng} determines the gross electric and recirculating powers directly:

$$P_{\text{et}} = P^*_{\text{net}} \cdot \frac{Q_{\text{eng}}}{Q_{\text{eng}} - 1}, \tag{103}$$

$$P_{\text{recirc}} = P_{\text{et}}/Q_{\text{eng}}. \tag{104}$$

The driver power follows from the recirculating budget. For thermal conversion, $P_{\text{recirc}} = P_{\text{drv}}/\eta_{\text{pin}} + P_{\text{fixed}}$; for inductive DEC, $P_{\text{recirc}} = P_{\text{drv}}(1/\eta_{\text{pin}} - 1) + P_{\text{fixed}}$, where P_{fixed} collects pumping, subsystem, auxiliary, cryogenic, target, and coil loads. Then $E_{\text{drv}} = P_{\text{drv}}/f_{\text{rep}}$, and P_{fus} is obtained from the forward energy balance with the derived E_{drv}. The inversion is closed-form (linear in P_{drv} and P_{fus}).

D.2 Steady-State Inverse

Substituting the forward power balance into eq. (30) gives $P_{\text{net}}(P_{\text{fus}})$; the inverse problem is to find the root $P_{\text{net}}(P_{\text{fus}}) = P^*_{\text{net}}$.

For fixed plasma parameters (n_e, T_e, B), the radiated power P_{rad} is independent of P_{fus}. However, the effective heating power (eq. (25)) introduces a nonlinearity: when $P_{\text{rad}} > P_{\text{cp}} + P_{\text{in}}$, the effective heating depends on $P_{\text{cp}} = f_{\text{cp}}\,P_{\text{fus}}$ (where f_{cp} is the charged-particle fraction), making $P_{\text{in,eff}}$ a function of P_{fus}. The net electric power is therefore piecewise linear in P_{fus}, with two regimes:

1. **Uncapped** ($P_{\text{rad}} < P_{\text{cp}} + P_{\text{in}}$): $P_{\text{in,eff}} = P_{\text{in}}$ (constant); the forward balance is linear in P_{fus} and admits a closed-form solution.
2. **Radiation-limited** ($P_{\text{rad}} \geq P_{\text{cp}} + P_{\text{in}}$): $P_{\text{in,eff}} = P_{\text{rad}} - f_{\text{cp}}\, P_{\text{fus}}$; the forward balance is linear with a different slope (increasing P_{fus} provides more ash power to offset radiation, reducing required heating).

The inverse solver uses Newton iteration with an analytical piecewise-constant Jacobian, initialized from the closed-form uncapped solution. Because $P_{\text{net}}(P_{\text{fus}})$ is piecewise linear, convergence is exact in at most two iterations. The Newton step is:

$$P_{\text{fus}}^{(k+1)} = P_{\text{fus}}^{(k)} - \frac{P_{\text{net}}(P_{\text{fus}}^{(k)}) - P_{\text{net}}^{*}}{\mathrm{d}P_{\text{net}}/\mathrm{d}P_{\text{fus}}\big|_{P_{\text{fus}}^{(k)}}} \tag{105}$$

where the derivative is computed analytically for each regime rather than via automatic differentiation, avoiding nested differentiation during sensitivity analysis.

E Reference Walkthrough Script

The full reference script for the section 2 walkthrough is reproduced below. It is the canonical 1 GWe D-^{3}He steady-state mirror reference cited in tables 1 and 2, and lives in the repository at `examples/external_physics_handoff.py`.

```
"""Canonical 1 GWe D-3He steady-state mirror reference for the 1costingfe paper.

This is the example cited in Section 2.2: physics outputs from an external
model (central-cell geometry, field, temperatures, densities, DEC fractions,
secondary burn fractions) hand off to the 1costingfe forward call, which
produces a complete CAS-account rollup and an LCOE figure.

The forward call uses an inverse power-balance solve: given the engineering
parameters (geometry, efficiencies, burn fractions) and the net electric
target, the model finds the required fusion power and derives all costs.

Numbers are illustrative; the point is the handoff pattern, not the design.
"""

from costingfe import ConfinementConcept, CostModel, Fuel, PowerCycle


def main() -> None:
    model = CostModel(
        concept=ConfinementConcept.MIRROR,
        fuel=Fuel.DHE3,
        power_cycle=PowerCycle.RANKINE,
    )

    result = model.forward(
        # Customer parameters
        net_electric_mw=1000.0,
        availability=0.87,  # mirror default (axial access shortens outages)
        lifetime_yr=30,
        construction_time_yr=6.0,
        interest_rate=0.07,
        inflation_rate=0.02,
        noak=True,
        # Geometry (Section 2.2 Table: central-cell length, plasma radius)
        R0=0.0,  # cylinder axis (no major-radius offset)
        chamber_length=80.0,  # m, central-cell length
        plasma_t=0.4,  # m, plasma radius at midplane
        elon=1.0,  # circular cross-section
        blanket_t=0.30,  # m, thin blanket (D-3He: low neutron fluence)
        ht_shield_t=0.20,  # m
        structure_t=0.15,  # m
```

```
    vessel_t=0.10,  # m
    # Magnets
    b_center=12.0,  # T, on-axis field at the coil center
    r_bore=1.85,  # m, effective winding bore radius
    # Power balance (Section 2.2 Table)
    p_input=50.0,  # MW, NBI heating (neutral beams sustain mirror)
    eta_couple=1.0,  # NBI coupling; eta_pin = 0.60 x 1.0 = 0.60
    eta_p=0.50,  # pumping efficiency
    p_coils=5.0,  # MW, solenoid + mirror coils
    p_cool=25.0,  # MW, first-wall cooling
    p_pump=1.5,  # MW
    p_house=4.0,  # MW
    p_cryo=1.0,  # MW
    f_sub=0.03,  # BOP subsystem fraction
    # Blanket / neutronics: D-3He is aneutronic primary, no blanket needed
    blanket_form="none",
    blanket_fill="none",
    mn=1.0,  # no neutron multiplication without a breeding blanket
    # Conversion: venetian-blind DEC on end-loss ions + thermal
    # (eta_th supplied by the RANKINE preset on CostModel)
    eta_de=0.70,  # venetian-blind DEC efficiency on end-loss ions
    f_dec=0.90,  # fraction of transport power routed to DEC
    # Plasma parameters for radiation calculation (Section 2.2 Table)
    n_e=3.3e19,  # m^-3, electron density
    T_e=70.0,  # keV, electron temperature
    Z_eff=1.3,  # effective ion charge
    B=3.0,  # T, central-cell field
    plasma_volume=400.0,  # m^3 (pi * plasma_t^2 * chamber_length)
    T_edge=0.20,  # keV, edge temperature (open field lines)
    tau_ratio=3.0,  # impurity confinement time / energy confinement time
    R_w=0.4,  # wall reflectivity (lower for open ends)
    # D-3He secondary burn fractions (Section 2.2 Table)
    dhe3_f_T=0.5,  # secondary D-T burn fraction
    dhe3_dd_frac=0.131,  # D-D side-reaction fraction
    dd_f_T=0.969,
    dd_f_He3=0.689,
)

costs = result.costs
pt = result.power_table

# Group-level rollup (matches paper Section 2.2 table)
# CAS20 = sum of CAS21-29 (direct capital); CAS30 is indirect, shown separately.
cas23_29 = (
    costs.cas23
    + costs.cas24
    + costs.cas25
    + costs.cas26
    + costs.cas27
    + costs.cas28
    + costs.cas29
)
rows = [
    ("CAS10", costs.cas10),
    ("CAS21", costs.cas21),
    ("CAS22", costs.cas22),
    ("CAS23-29", cas23_29),
    ("CAS30", costs.cas30),
    ("CAS40", costs.cas40),
    ("CAS50", costs.cas50),
    ("CAS60", costs.cas60),
    ("Total overnight", costs.total_capital),
    ("CAS70 (M$/yr)", costs.cas70),
    ("CAS80 (M$/yr)", costs.cas80),
    ("CAS90 (M$/yr)", costs.cas90),
```

```
        ("LCOE", costs.lcoe),
    ]
    print("1 GWe D-3He mirror with venetian-blind DEC, NOAK reference\n")
    for label, value in rows:
        print(f"  {label:<25s} {float(value):>10.1f}")
    print()
    print("Power balance:")
    print(f"  P_fus:   {pt.p_fus:.0f} MW")
    print(f"  P_net:   {pt.p_net:.0f} MW")
    print(f"  Q_eng:   {pt.q_eng:.2f}")
    print(f"  Recirc:  {pt.rec_frac:.1%}")
    print()
    print(f"Overnight specific cost: {costs.overnight_cost:.0f} $/kW")
    print(f"LCOE: {costs.lcoe:.1f} $/MWh")


if __name__ == "__main__":
    main()
```

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